

INTERNATIONAL STANDARD

**Surface mounted piezoelectric devices for frequency control and selection –
Standard outlines and terminal lead connections –
Part 2: Ceramic enclosures**



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INTERNATIONAL STANDARD

**Surface mounted piezoelectric devices for frequency control and selection –
Standard outlines and terminal lead connections –
Part 2: Ceramic enclosures**

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**SURFACE MOUNTED PIEZOELECTRIC DEVICES FOR
FREQUENCY CONTROL AND SELECTION – STANDARD
OUTLINES AND TERMINAL LEAD CONNECTIONS –****Part 2: Ceramic enclosures****FOREWORD**

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International Standard IEC 61837-2 has been prepared by IEC technical committee 49: Piezoelectric, dielectric and electrostatic devices and associated materials for frequency control, selection and detection.

This third edition cancels and replaces the second edition published in 2011 and Amendment 1:2014. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

- a) revision of the figures to match the notation of the drawings of IEC 61240:2016;
- b) addition of 7 enclosures as follows: DCC-6/5032A, DCC-6/3225A, DCC-4/3215C, DCC-6/2016A, DCC-2/2012C, DCC-2/1610C, DCC-4/1210C.

As a result, this third edition contains a total of 45 enclosure types, which are listed in Table 1.

This International Standard is to be read in conjunction with IEC 61240:2016.

The text of this International Standard is based on the following documents:

CDV	Report on voting
49/1252/CDV	49/1276/RVC

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 61837 series, published under the general title *Surface mounted piezoelectric devices for frequency control and selection – Standard outlines and terminal lead connections*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

SURFACE MOUNTED PIEZOELECTRIC DEVICES FOR FREQUENCY CONTROL AND SELECTION – STANDARD OUTLINES AND TERMINAL LEAD CONNECTIONS –

Part 2: Ceramic enclosures

1 Scope

This part of IEC 61837 deals with standard outlines and terminal lead connections as they apply to surface-mounted devices (SMD) for frequency control and selection in ceramic enclosures, and is based on IEC 61240:2016.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61240:2016, *Piezoelectric devices – Preparation of outline drawings of surface-mounted devices (SMD) for frequency control and selection – General rules*

3 Terms and definitions

No terms and definitions are listed in this document.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

4 Configuration of enclosures

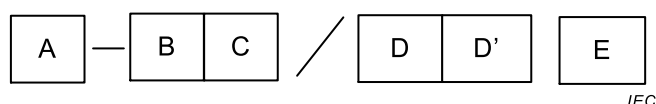
The enclosures of the surface-mounted devices are made of ceramic materials with the terminals of deposited metal film (leadless type) based on a descriptive designation system for semiconductors – devices packages.

The configuration symbols are shown as follows:

- DCC (dual chip carrier);
- QCC (quad chip carrier).

5 Designation of types

The designation of types is shown on four parts as follows:



A: Configuration symbol of enclosures:

- DCC (dual chip carrier);
- QCC (quad chip carrier).

B: Structure of terminal leads: leadless type has no mark.

C: Number of terminal leads

D: Serial number of both figures

E: Arrangement of terminal land:

In the case of DCC types:

- A (arrangement in the width direction side or the width direction side and the corner);
- B (arrangement in the length direction side or the length direction side and the corner);
- C (arrangement in the corner only).

In the case of QCC types:

- A (arrangement in the side only);
- B (arrangement in the side and the corner).

6 Ceramic enclosure dimensions

The dimensions given in this document apply to all completed SMD-devices for frequency control and selection. Only those dimensions which meet the requirements of IEC 61240 are given.

7 Lead connections

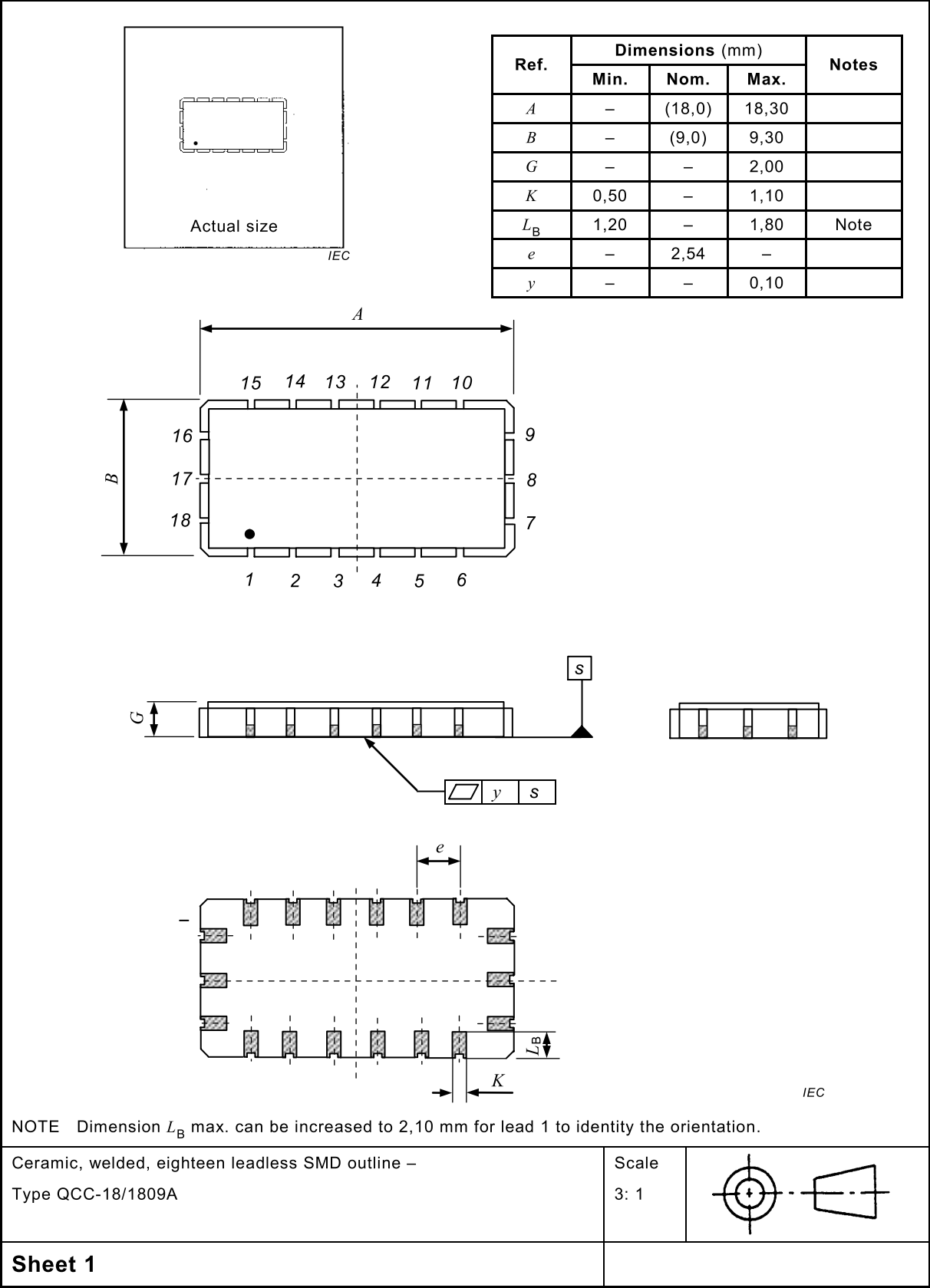
Recommendations for the lead connections of all completed SMD-devices for frequency control and selection are given in the following individual sheets. Lead connections shall always be given in the detail specification.

8 Designation of ceramic enclosures

Table 1 is a list which includes all new enclosure types with their sheet numbers and brief descriptions. Old enclosure names are also listed as references.

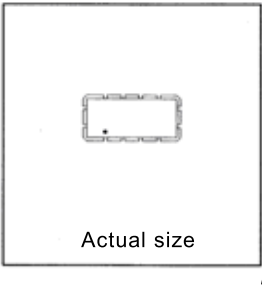
Table 1 – Designation of ceramic enclosures

No.	Type	Old type	Sheet No.	Description
1	QCC-18/1809A	QCC-18/01	Sheet 1	Ceramic, welded, eighteen leadless SMD outline
2	QCC-12/1407A		Sheet 2	Ceramic, welded, twelve leadless SMD outline
3	DCC-4/1206A	DCC-4/01	Sheet 3	Ceramic, four leadless SMD outline
4	DCC-2/1206A		Sheet 4	Ceramic, two leadless SMD outline
5	QCC-10/9272A		Sheet 5	Ceramic, welded, ten leadless SMD outline
6	DCC-4/9070A		Sheet 6	Ceramic, welded, four leadless SMD outline
7	DCC-4/8045B	DCC-04/02, 03	Sheet 7	Ceramic, welded, four leadless SMD outline
8	DCC-2/8045B		Sheet 8	Ceramic, welded, two leadless SMD outline
9	DCC-6/7834B		Sheet 9	Ceramic, welded, six leadless SMD outline
10	DCC-6/7050A		Sheet 10	Ceramic, welded, six leadless SMD outline
11	DCC-4/7050A	DCC-4/08	Sheet 11	Ceramic, welded, four leadless SMD outline
12	DCC-4/7050B	DCC-4/04, 05	Sheet 12	Ceramic, welded, four leadless SMD outline
13	QCC-10/7050A		Sheet 13	Ceramic, welded, ten leadless SMD outline
14	QCC-6/7050A	QCC-6/01, 02	Sheet 14	Ceramic, welded, six leadless SMD outline
15	DCC-6/6035A	DCC-4/06, 07	Sheet 15	Ceramic, welded, six leadless SMD outline
16	DCC-4/6035C		Sheet 16	Ceramic, welded, four leadless SMD outline
17	DCC-2/6035C		Sheet 17	Ceramic, welded, two leadless SMD outline
18	QCC-8/5050A	QCC-8/02	Sheet 18	Ceramic, welded, eight leadless SMD outline
19	QCC-12/5045A	QCC-12/02	Sheet 19	Ceramic, welded, twelve leadless SMD outline
20	QCC-8/5045A		Sheet 20	Ceramic, welded, eight leadless SMD outline
21	DCC-6/5032A		Sheet 21	Ceramic, welded, six leadless SMD outline
22	DCC-4/5032A		Sheet 22	Ceramic, welded, four leadless SMD outline
23	DCC-4/5032C		Sheet 23	Ceramic, welded, four leadless SMD outline
24	DCC-2/5032B	DCC-2/01	Sheet 24	Ceramic, welded, two leadless SMD outline
25	DCC-2/4818C		Sheet 25	Ceramic, welded, two leadless SMD outline
26	DCC-2/4115C		Sheet 26	Ceramic, welded, two leadless SMD outline
27	DCC-4/4025C	DCC-6/01	Sheet 27	Ceramic, welded, four leadless SMD outline
28	QCC-8/3838A		Sheet 28	Ceramic, welded, eight leadless SMD outline
29	DCC-6/3838A		Sheet 29	Ceramic, welded, six leadless SMD outline
30	DCC-6/3225A		Sheet 30	Ceramic, welded, six leadless SMD outline
31	DCC-4/3225C		Sheet 31	Ceramic, welded, four leadless SMD outline
32	DCC-4/3215C		Sheet 32	Ceramic, welded, four leadless SMD outline
33	DCC-2/3215C		Sheet 33	Ceramic, welded, two leadless SMD outline
34	QCC-8/3030B		Sheet 34	Ceramic, welded, eight leadless SMD outline
35	DCC-6/3030A		Sheet 35	Ceramic, welded, six leadless SMD outline
36	DCC-6/2520A		Sheet 36	Ceramic, welded, six leadless SMD outline
37	DCC-4/2520C/01,02,03		Sheet 37	Ceramic, welded, four leadless SMD outline
38	DCC-4/2020C		Sheet 38	Ceramic, welded, six leadless SMD outline
39	DCC-6/2016A		Sheet 39	Ceramic, welded, four leadless SMD outline
40	DCC-4/2016C/01,02,03		Sheet 40	Ceramic, welded, four leadless SMD outline
41	DCC-2/2012C		Sheet 41	Ceramic, welded, two leadless SMD outline
42	DCC-4/1612C/01,02		Sheet 42	Ceramic, welded, four leadless SMD outline
43	DCC-2/1612C		Sheet 43	Ceramic, welded, two leadless SMD outline
44	DCC-2/1610C		Sheet 44	Ceramic, welded, two leadless SMD outline
45	DCC-4/1210C		Sheet 45	Ceramic, welded, four leadless SMD outline



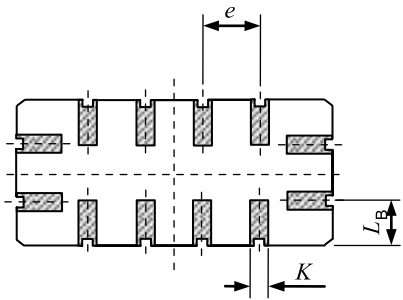
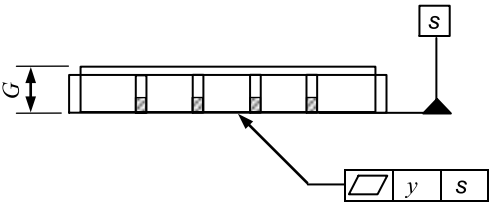
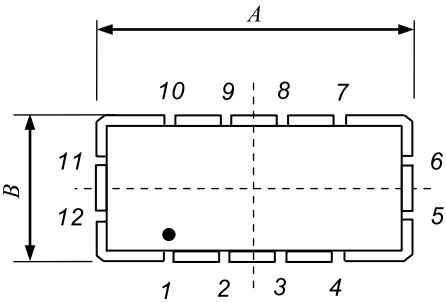
Terminal land connections of Type QCC-18/1809A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1				Ground
2				Option
3				Option
4				Option
5				Option
6				Ground
7				Input/Output
8				Ground
9				Input/Output /Ground
10				Ground
11				Option
12				Option
13				Option
14				Option
15				Ground
16				Output/Input
17				Ground
18				Output/Input /Ground



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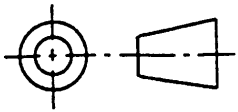
Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
<i>A</i>	–	(14,0)	14,30	
<i>B</i>	–	(6,5)	6,80	
<i>G</i>	–	–	2,00	
<i>K</i>	0,50	–	1,10	
<i>L_B</i>	1,70	–	2,30	
<i>e</i>	–	2,54	–	
<i>y</i>	–	–	0,10	



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Ceramic, welded, twelve leadless SMD outline –
Type QCC-12/1407A

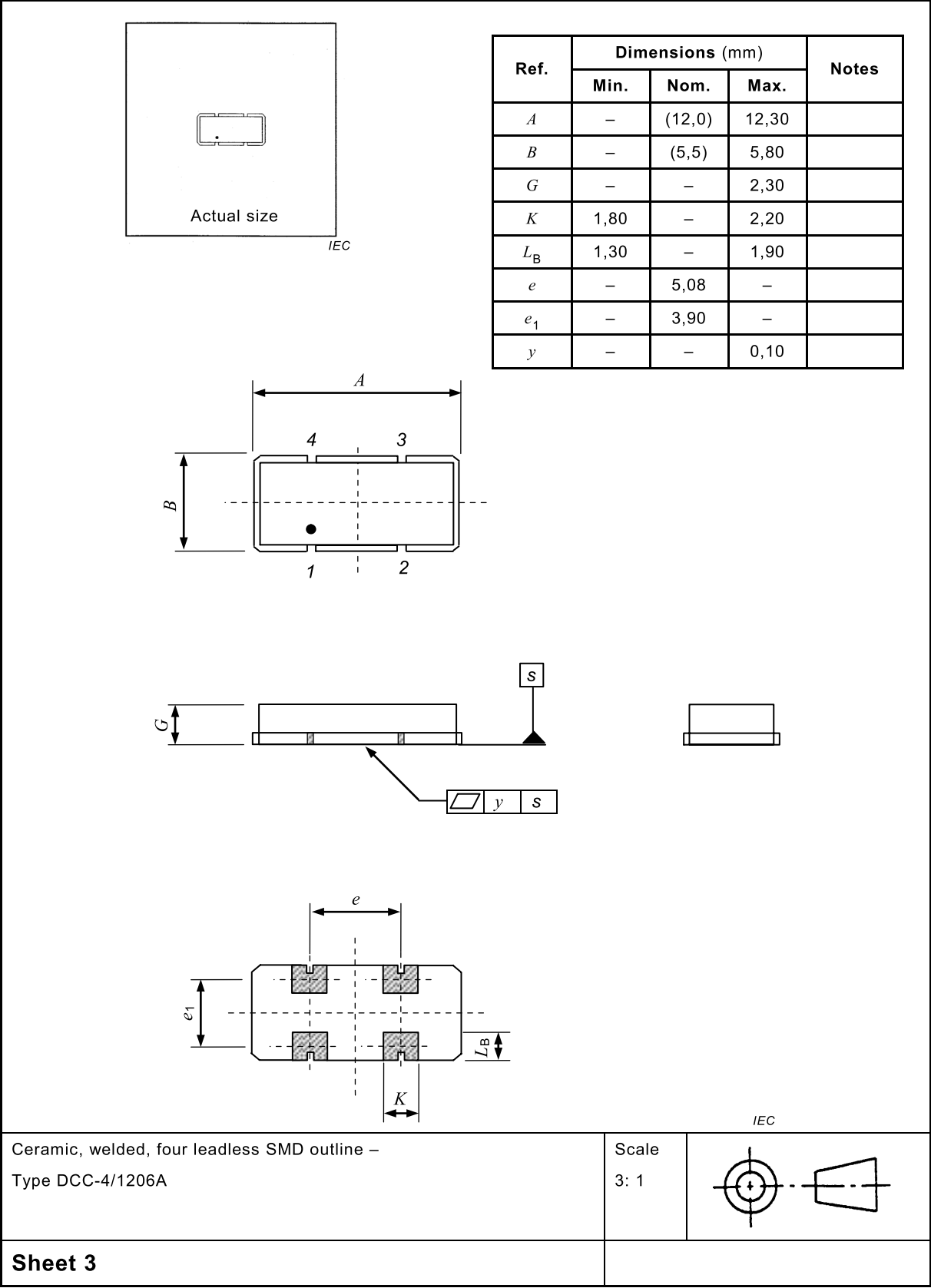
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Sheet 2

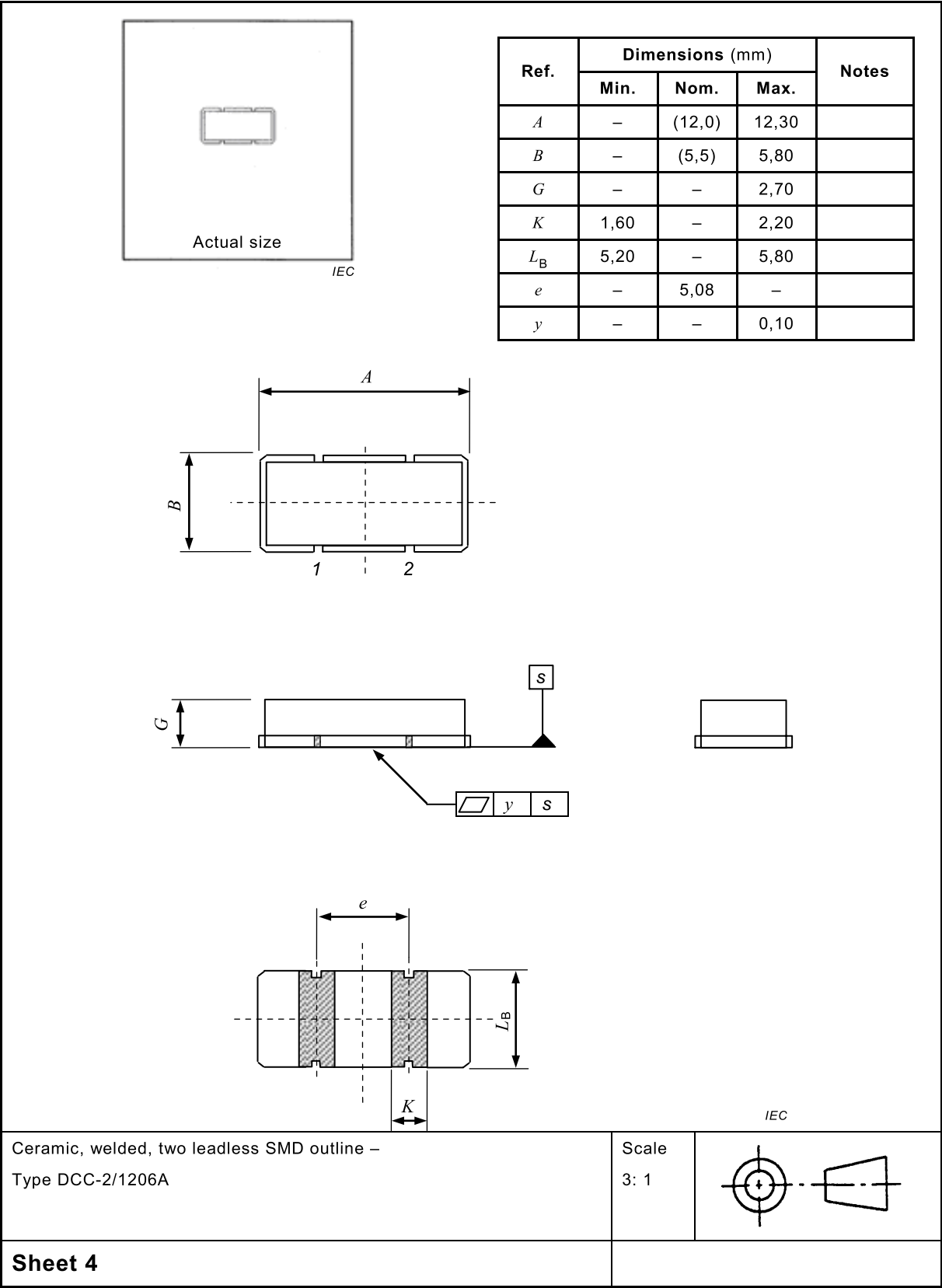
Terminal land connections of Type QCC-12/1407A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1				Option
2				Option
3				Ground
4				Option
5				Output/Input
6				Output/Input /Ground
7				Option
8				Ground
9				Option
10				Option
11				Input/Output
12				Input/Output /Ground



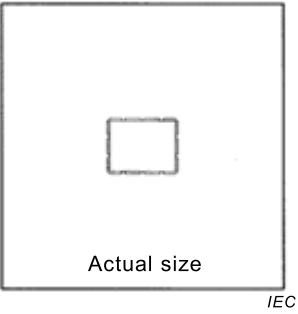
Terminal land connections of Type DCC-4/1206A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1	Control voltage		
2	NC	Ground		
3	Terminal 2	Output		
4	NC	DC supply		

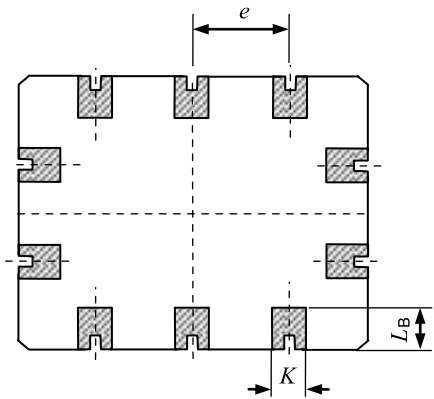
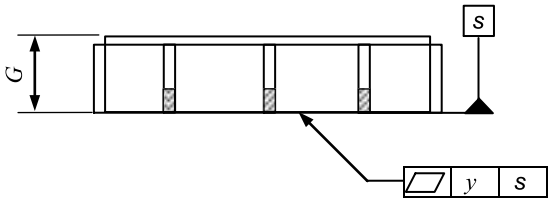
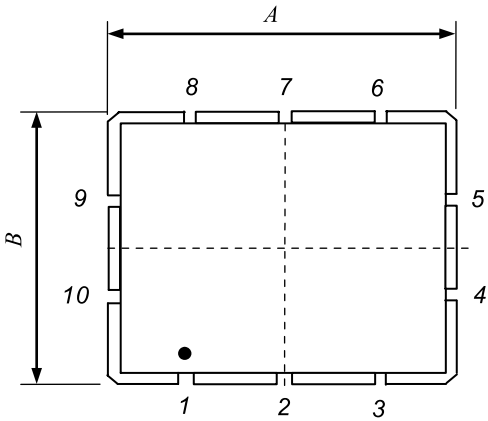


Terminal land connections of Type DCC-2/1206A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			



Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
<i>A</i>	–	(9,2)	9,50	
<i>B</i>	–	(7,2)	7,50	
<i>G</i>	–	–	2,00	
<i>K</i>	0,60	–	1,20	
<i>L_B</i>	0,80	–	1,40	Note
<i>e</i>	–	2,54	–	
<i>y</i>	–	–	0,10	

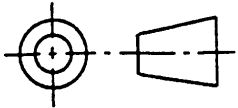


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NOTE Dimension *L_B* max. can be increased to 2,00 mm for lead 1 to identify the orientation.

Ceramic, welded, ten leadless SMD outline –
Type QCC-10/9272A

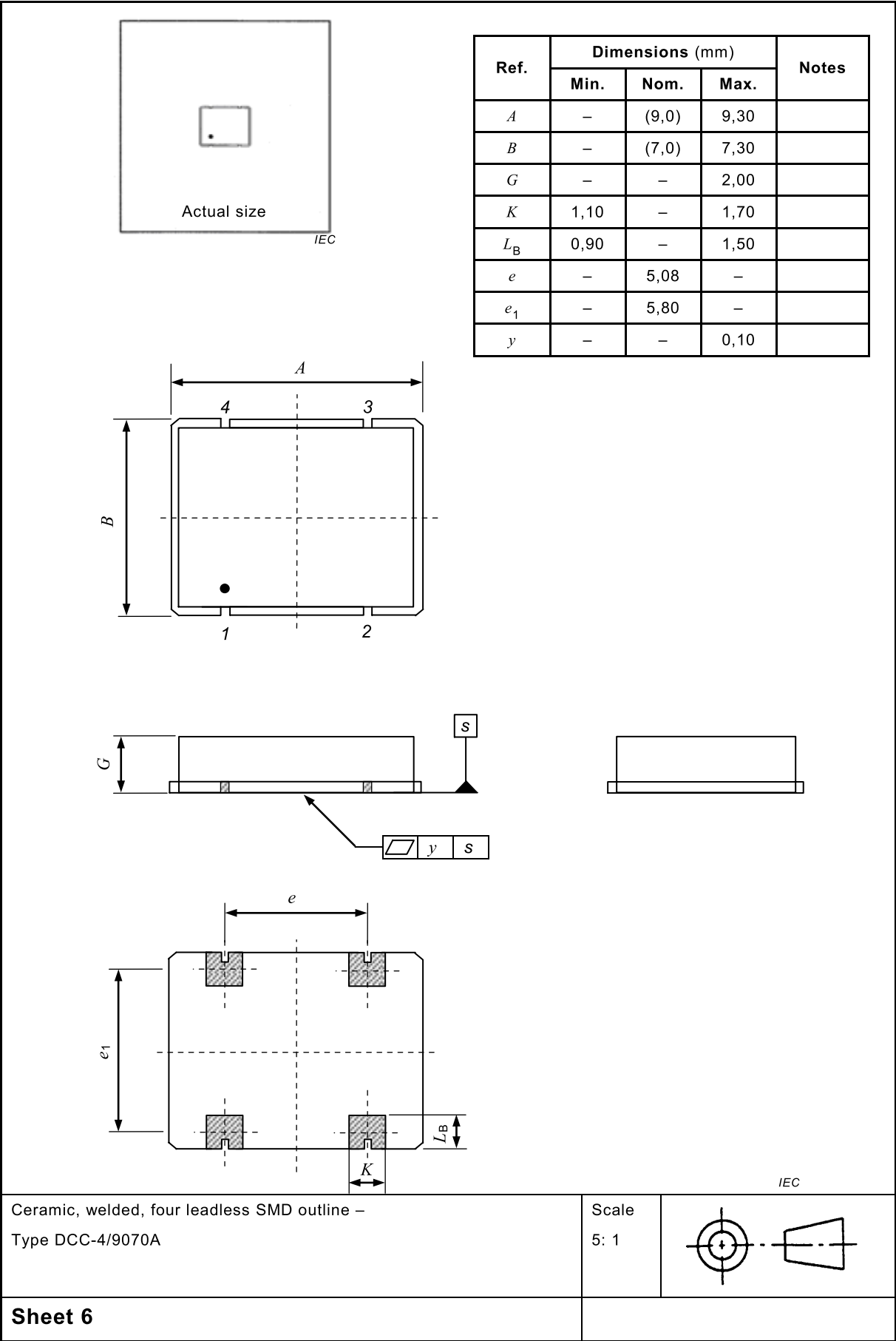
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Sheet 5

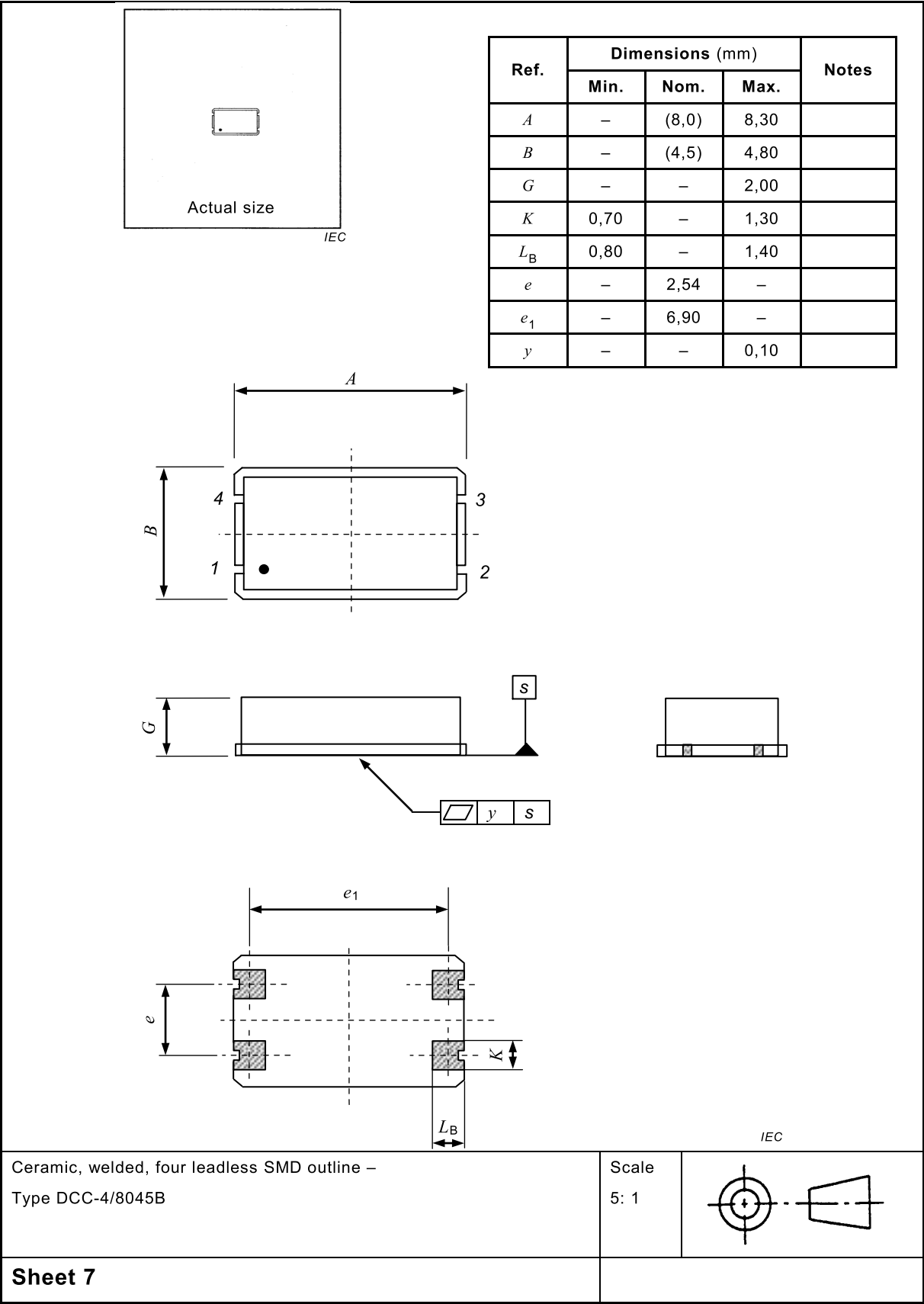
Terminal land connections of Type QCC-10/9272A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1				Input/Output /Ground
2				Input/Output
3				Ground
4				Ground
5				Ground
6				Output/Input /Ground
7				Output/Input
8				Option
9				Ground
10				Ground



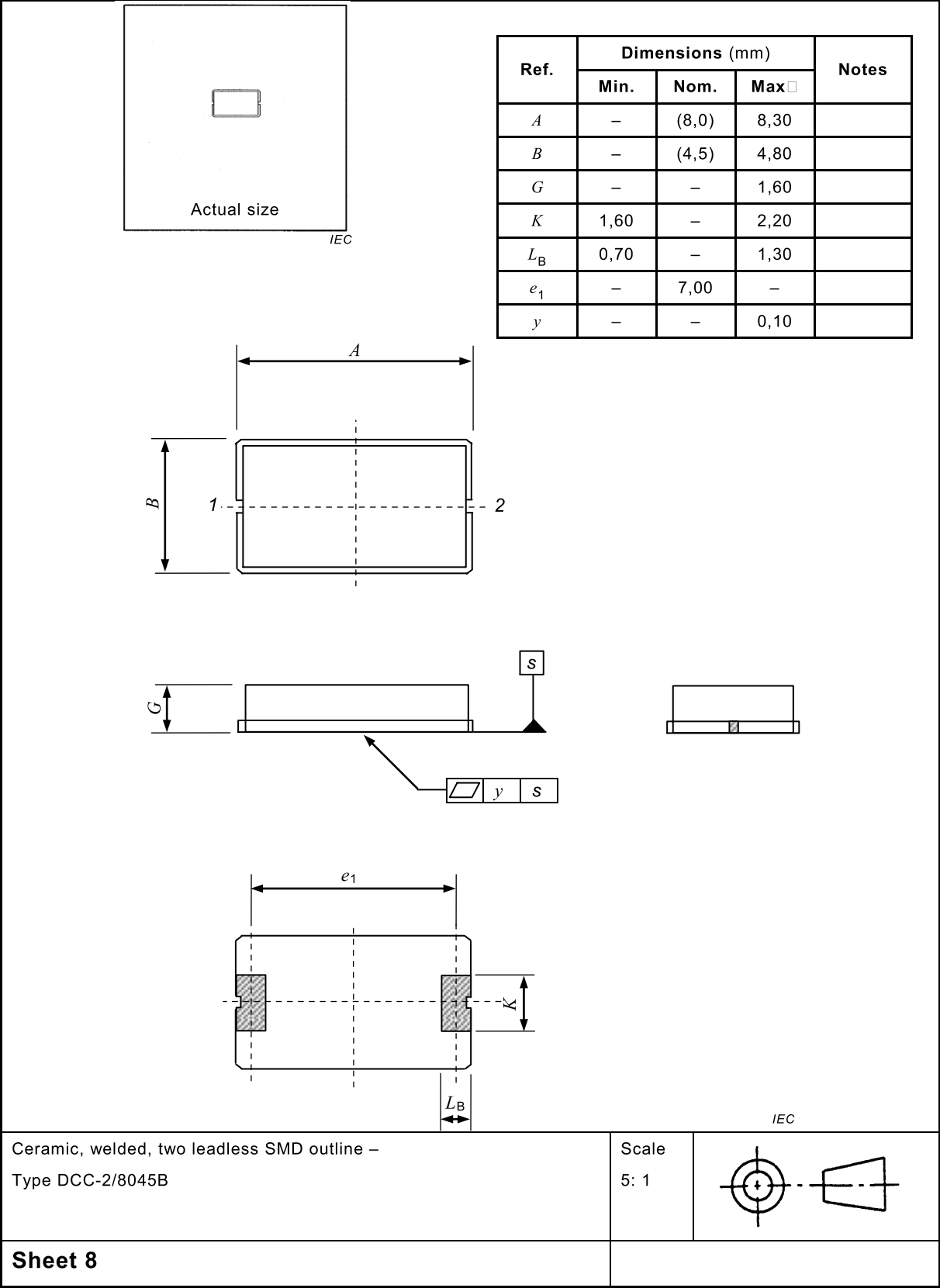
Terminal land connections of Type DCC-4/9070A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Control voltage		
2		Ground		
3		Output		
4		DC supply		



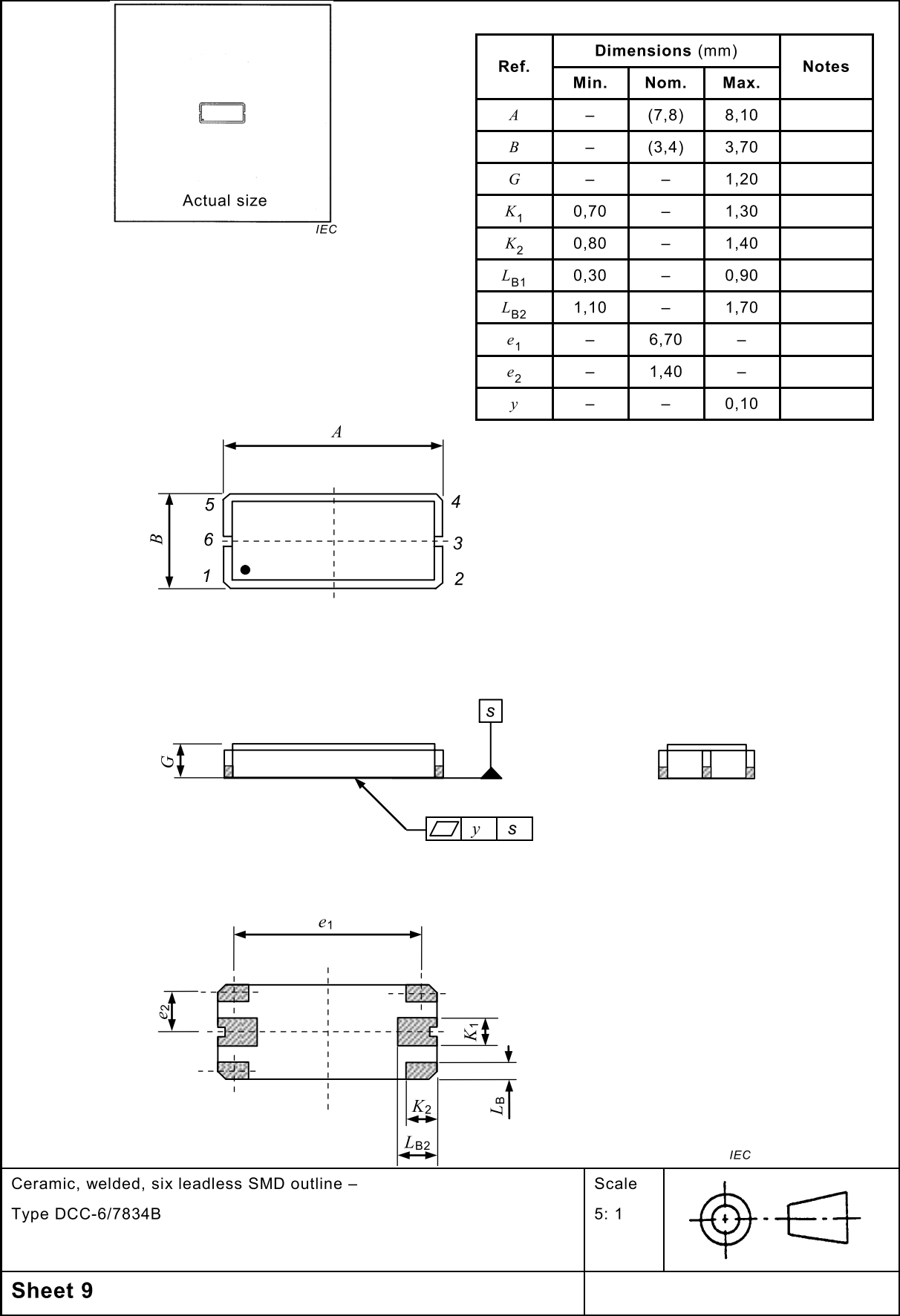
Terminal land connections of Type DCC-4/8045B

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1	Control voltage		
2	Option	Ground		
3	Terminal 2	Output		
4	Option	DC supply		



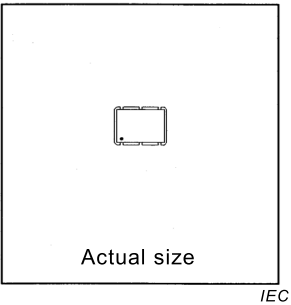
Terminal land connections of Type DCC-2/8045B

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			

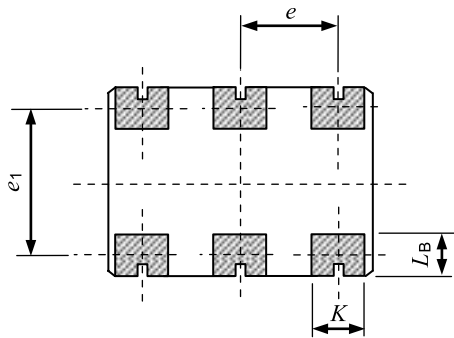
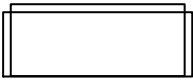
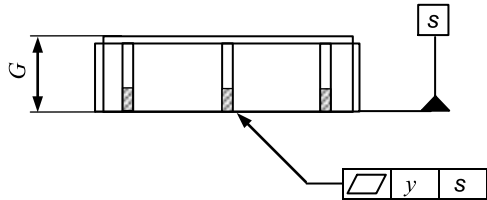
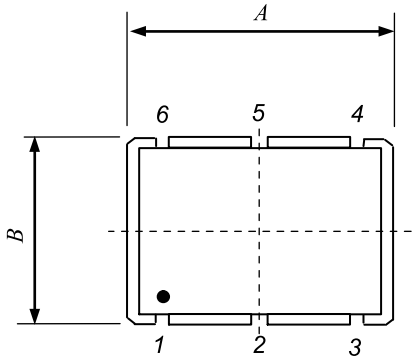


Terminal land connections of Type DCC-6/7834B

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Option			
2	Option			
3	Terminal 1			
4	Option			
5	Option			
6	Terminal 2			



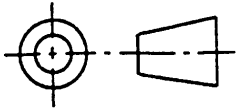
Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
<i>A</i>	–	(7,0)	7,30	
<i>B</i>	–	(5,0)	5,30	
<i>G</i>	–	–	2,00	
<i>K</i>	1,10	–	1,70	
<i>L_B</i>	0,80	–	1,40	
<i>e</i>	–	2,54	–	
<i>e₁</i>	–	3,90	–	
<i>y</i>	–	–	0,10	



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Ceramic, welded, six leadless SMD outline –
Type DCC-6/7050A

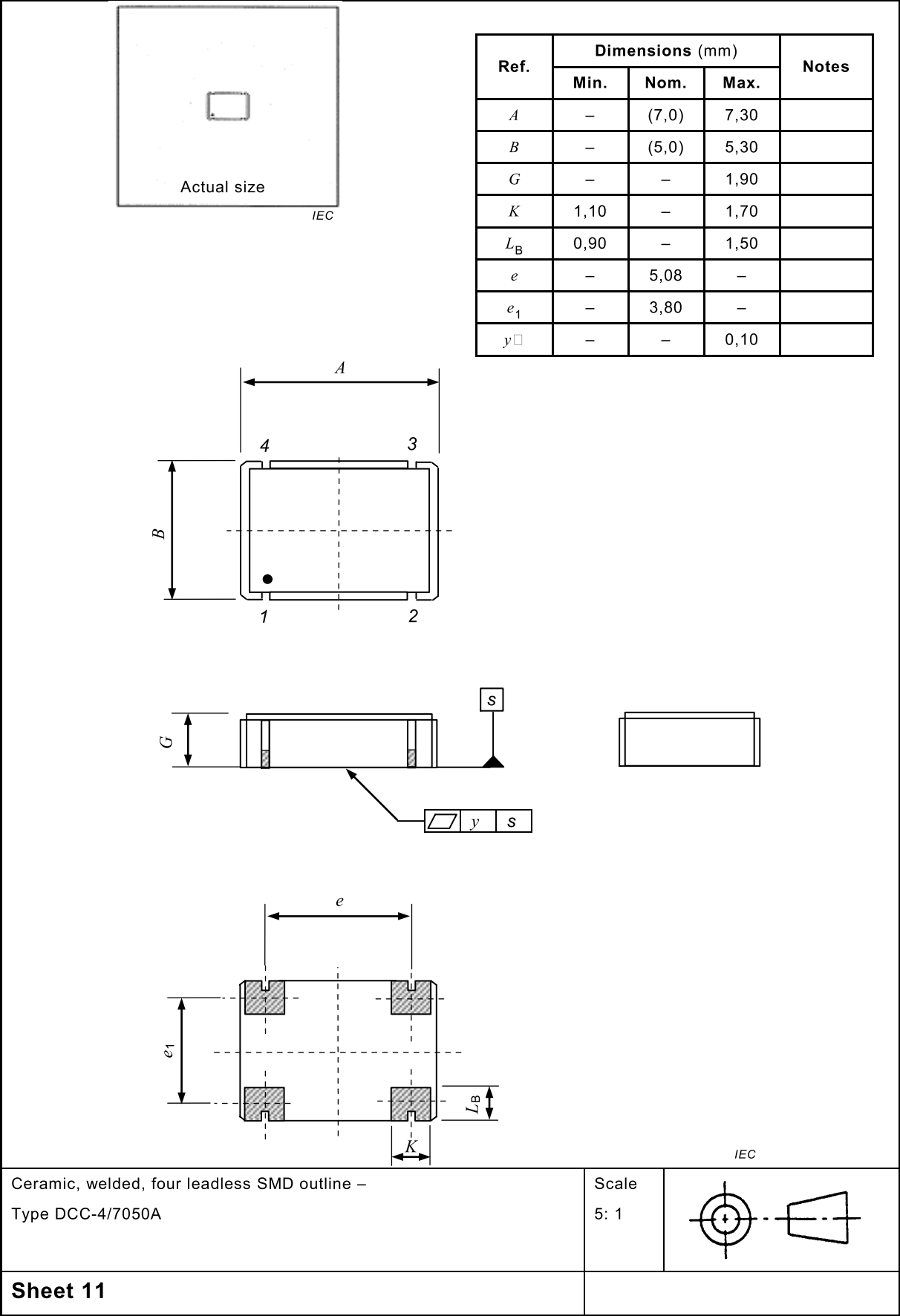
Scale
5: 1



Sheet 10

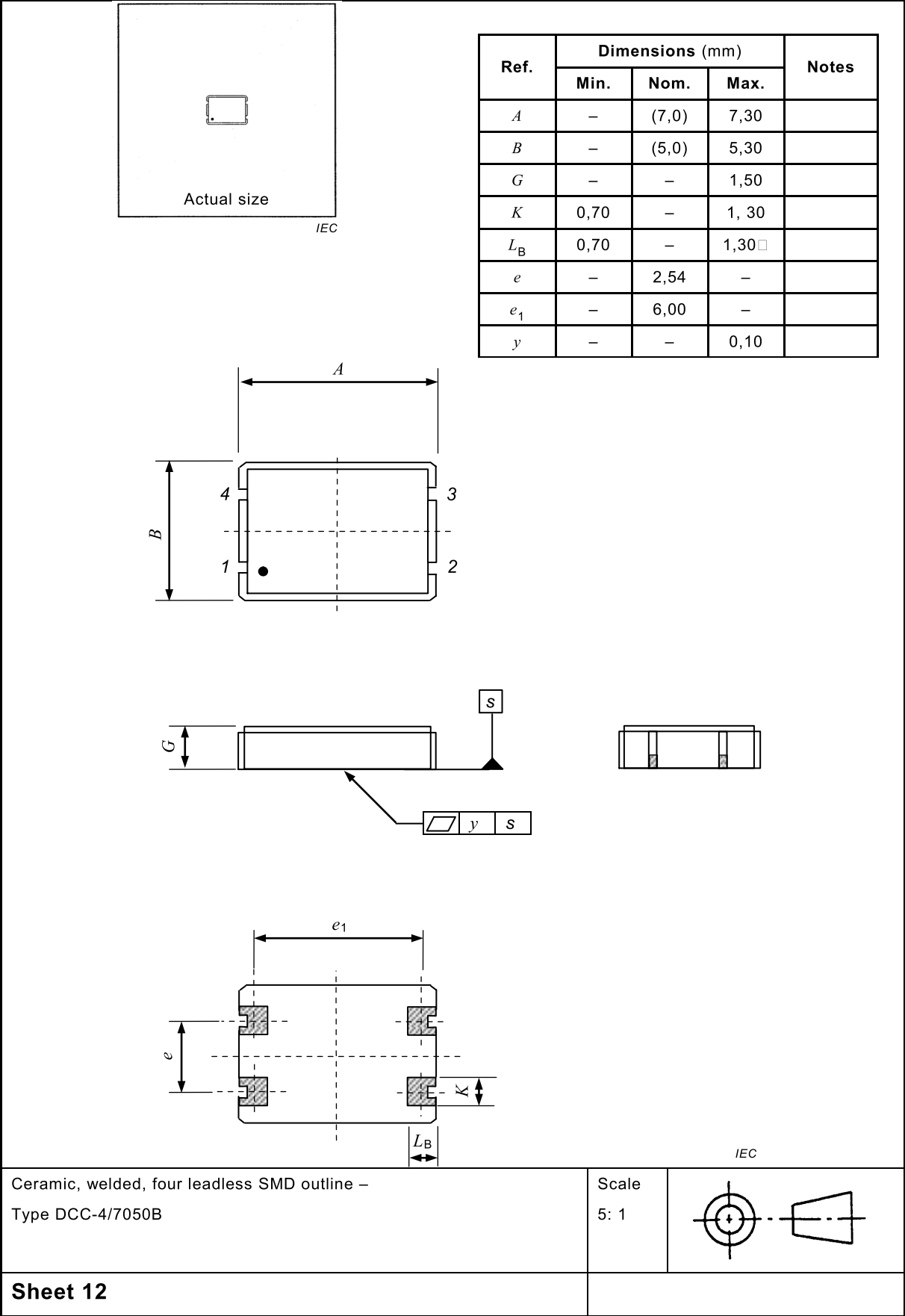
Terminal land connections of Type DCC-6/7050A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Output enable		
2		NC		
3		Ground		
4		Output		
5		Output		
6		DC supply		



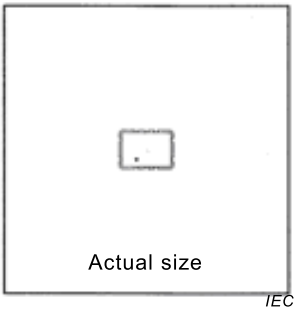
Terminal land connections of Type DCC-4/7050A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Control voltage		
2		Ground		
3		Output		
4		DC supply		

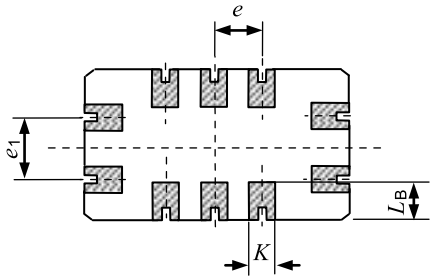
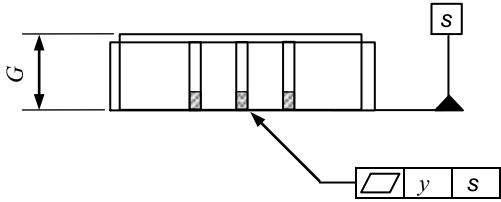
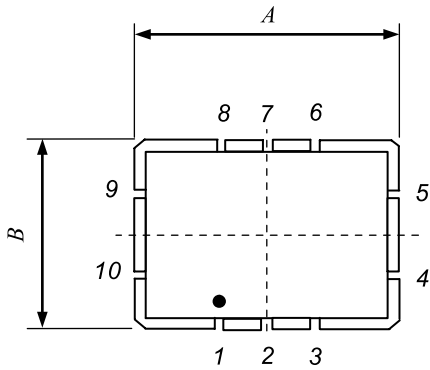


Terminal land connections of Type DCC-4/7050B

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Option			
3	Terminal 2			
4	Option			



Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
<i>A</i>	–	(7,0)	7,30	
<i>B</i>	–	(5,0)	5,30	
<i>G</i>	–	–	2,00	
<i>K</i>	0,40	–	1,00	
<i>L_B</i>	0,80	–	1,20	Note
<i>e</i>	–	1,27	–	
<i>e₁</i>	–	4,00	–	
<i>y</i>	–	–	0,10	

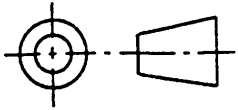


IEC

NOTE Dimension *L_B* max. can be increased to 1,60 mm for lead 1 to identify the orientation.

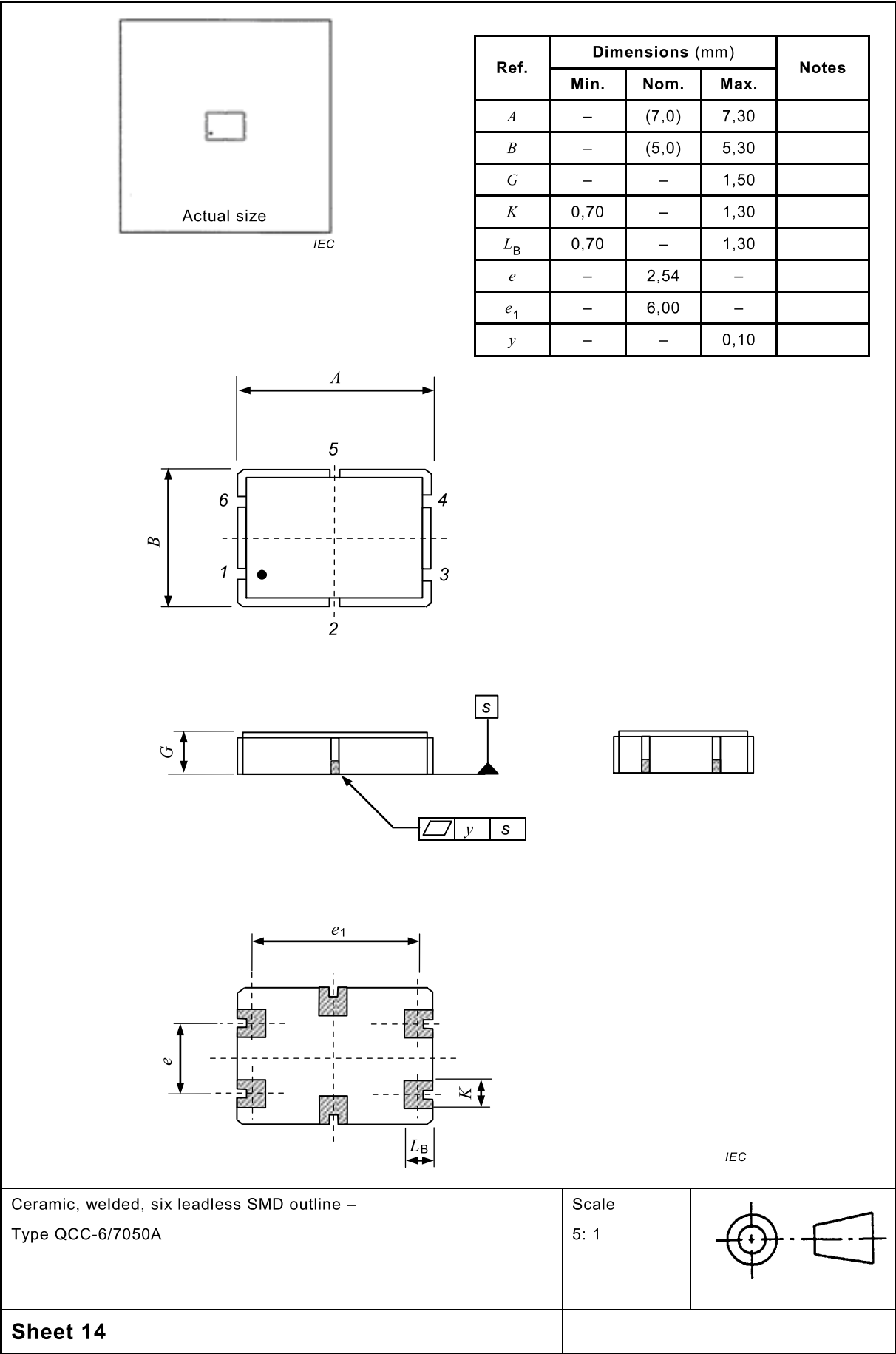
Ceramic, welded, ten leadless SMD outline –
Type QCC-10/7050A

Scale
5: 1



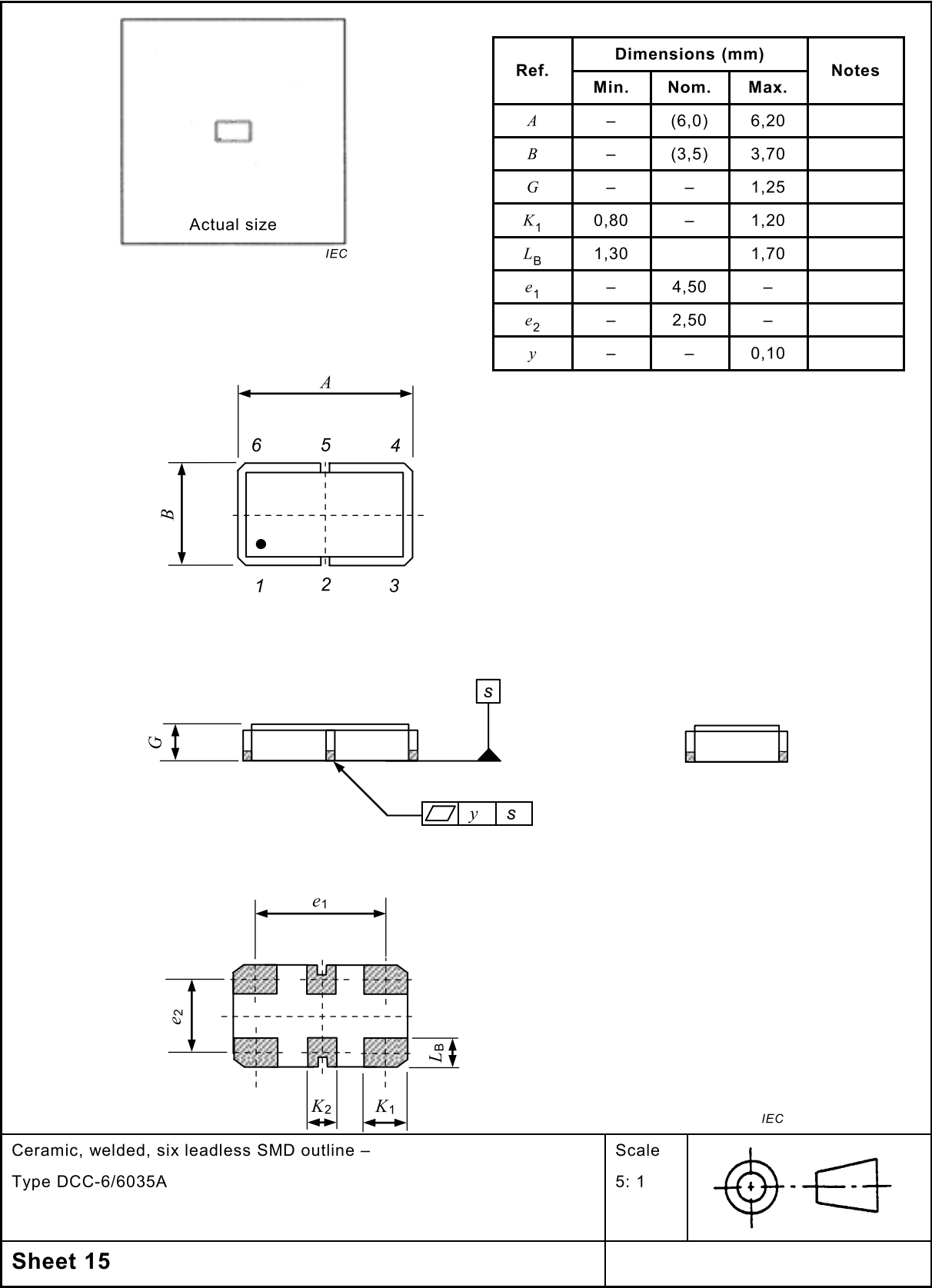
Terminal land connections of Type QCC-10/7050A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Option		Option
2		Option		Output/Input
3		Option		Option
4		Ground		Option
5		Output		Option
6		Option		Option
7		Option		Input/Output
8		Option		Option
9		DC supply		Ground
10		Option/Control voltage		Ground



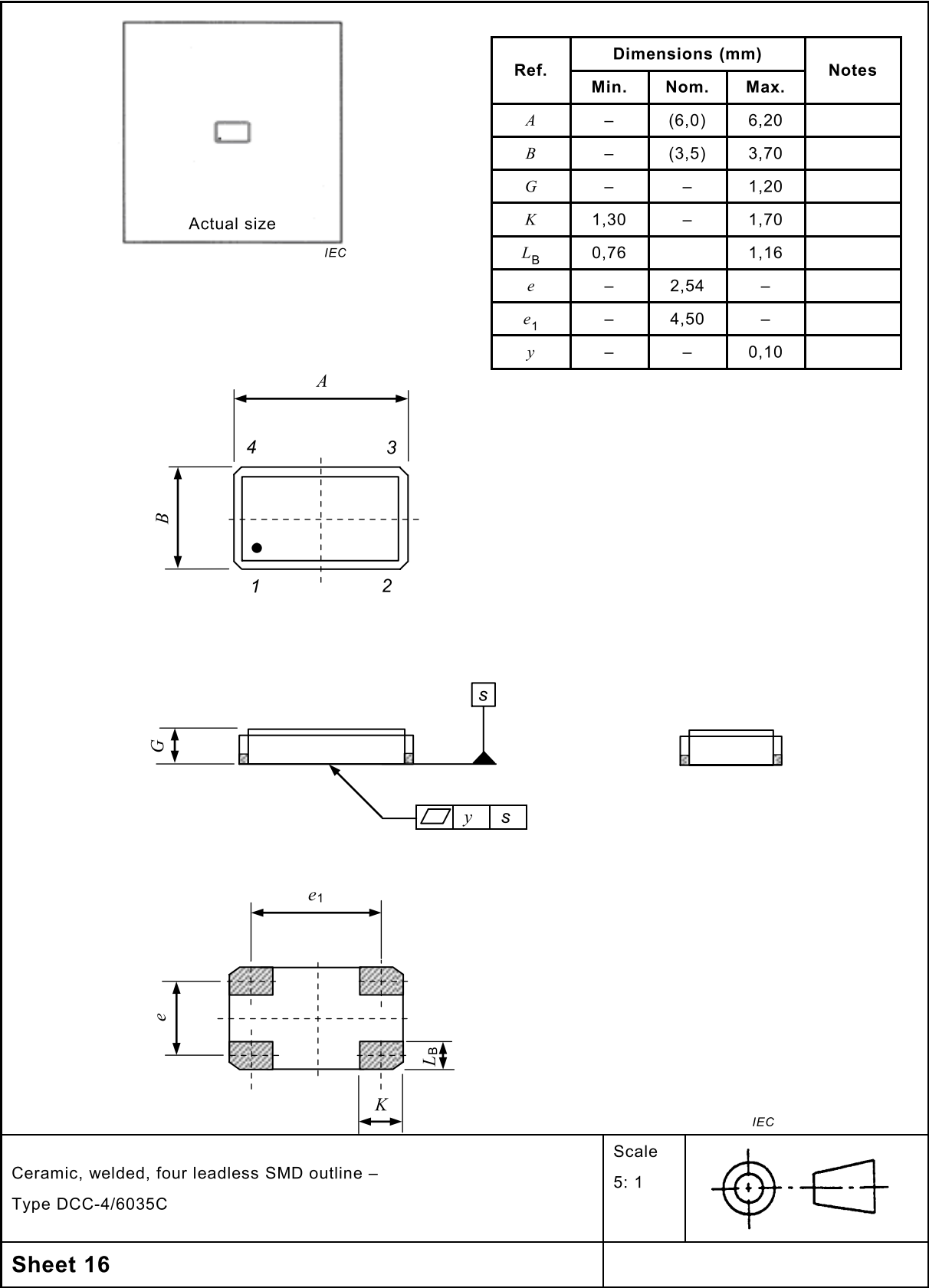
Terminal land connections of Type QCC-6/7050A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1			Ground	
2			Option	
3			Output/Input	
4			Ground	
5			Option	
6			Input/Output	



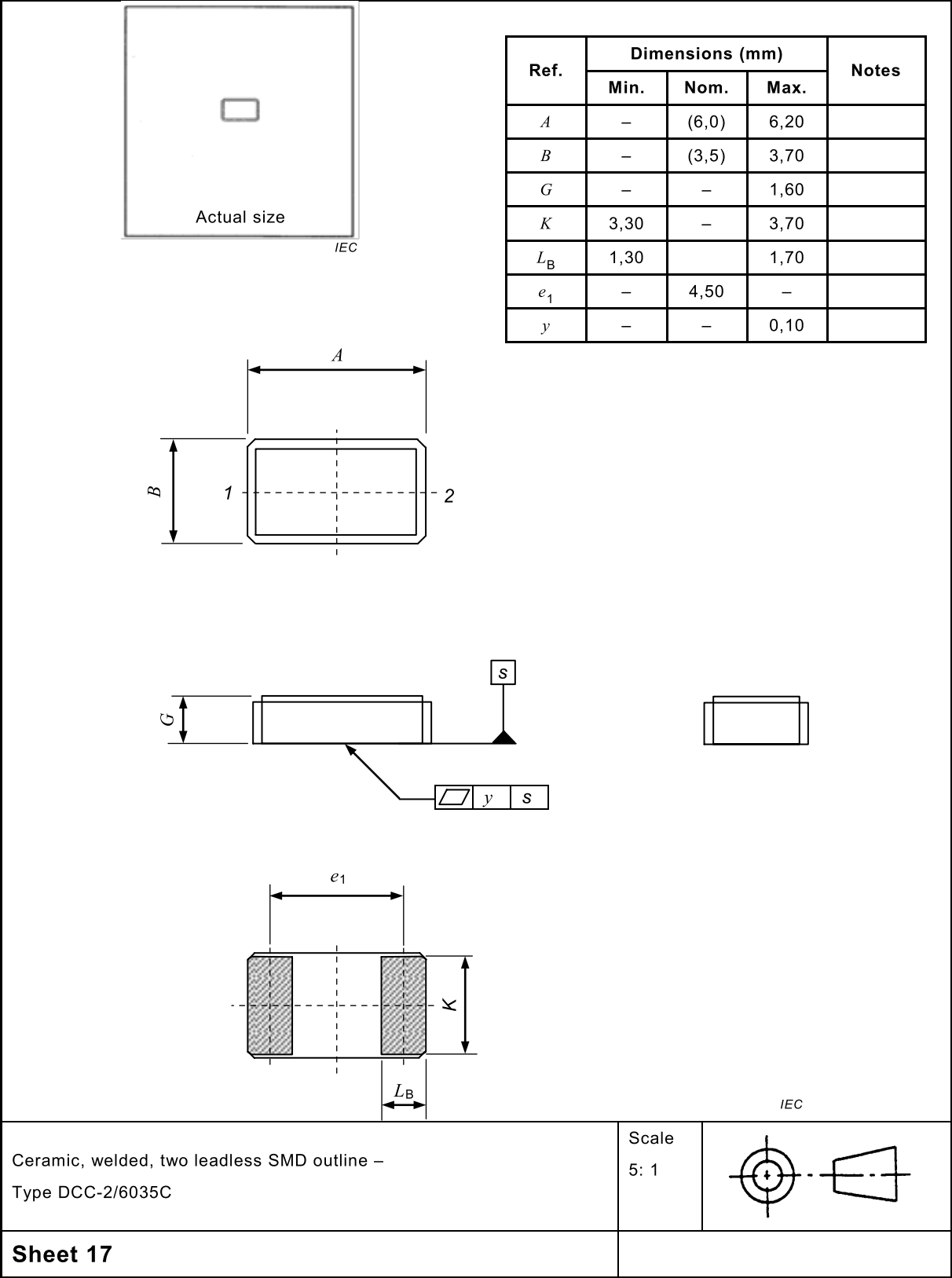
Terminal land connections of Type DCC-6/6035A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1			Ground	Ground
2			Ground	Input/Output
3			Output/Input	Ground
4			Ground	Ground
5			Ground	Output/Input
6			Input/Output	Ground



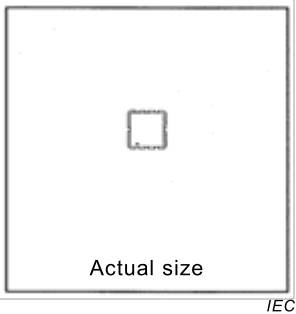
Terminal land connections of Type DCC-4/6035C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1	Control voltage		
2	Option	Ground		
3	Terminal 2	Output		
4	Option	DC supply		

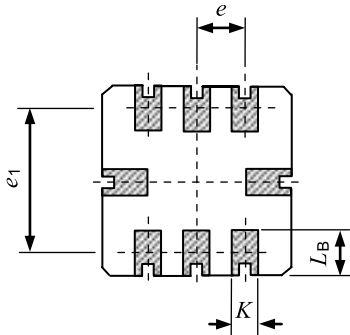
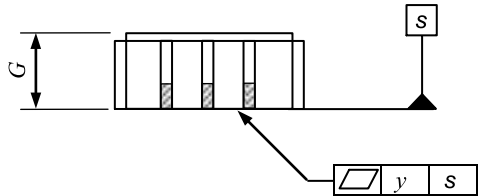
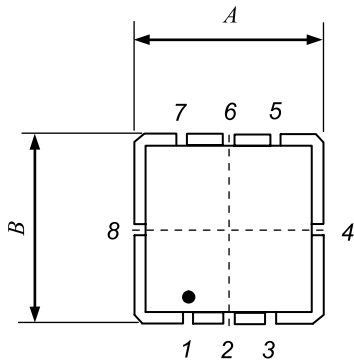


Terminal land connections of Type DCC-2/6035C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			



Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
<i>A</i>	–	(5,0)	5,20	
<i>B</i>	–	(5,0)	5,20	
<i>G</i>	–	–	2,00	
<i>K</i>	0,50	–	0,90	
<i>L_B</i>	1,00	–	1,40	Note
<i>e</i>	–	1,27	–	
<i>e₁</i>	–	3,80	–	
<i>y</i>	–	–	0,10	

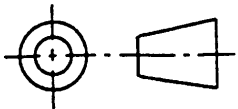


IEC

NOTE Dimension *L_B* max. can be increased to 1,90 mm for lead 8 to identify the orientation.

Ceramic, welded, eight leadless SMD outline –
Type QCC-8/5050A

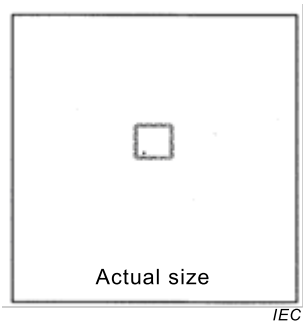
Scale
5: 1



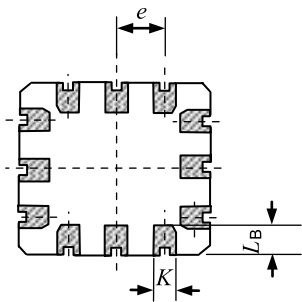
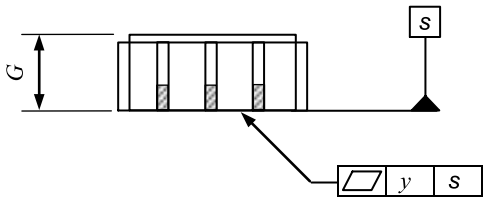
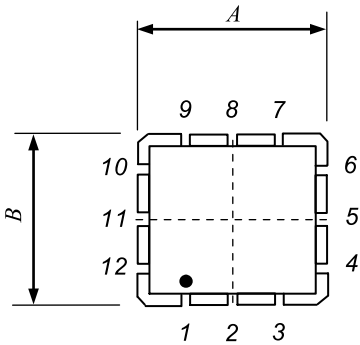
Sheet 18

Terminal land connections of Type QCC-8/5050A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1				Ground
2				Input/Output
3				Ground
4				Ground
5				Ground
6				Output/Input
7				Ground
8				Ground



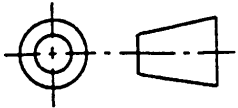
Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
<i>A</i>	–	(5,0)	5,20	
<i>B</i>	–	(4,5)	4,70	
<i>G</i>	–	–	2,00	
<i>K</i>	0,40	–	0,80	
<i>L_B</i>	0,60	–	1,00	
<i>e</i>	–	1,27	–	
<i>y</i>	–	–	0,10	



IEC

Ceramic, welded, twelve leadless SMD outline –
Type QCC-12/5045A

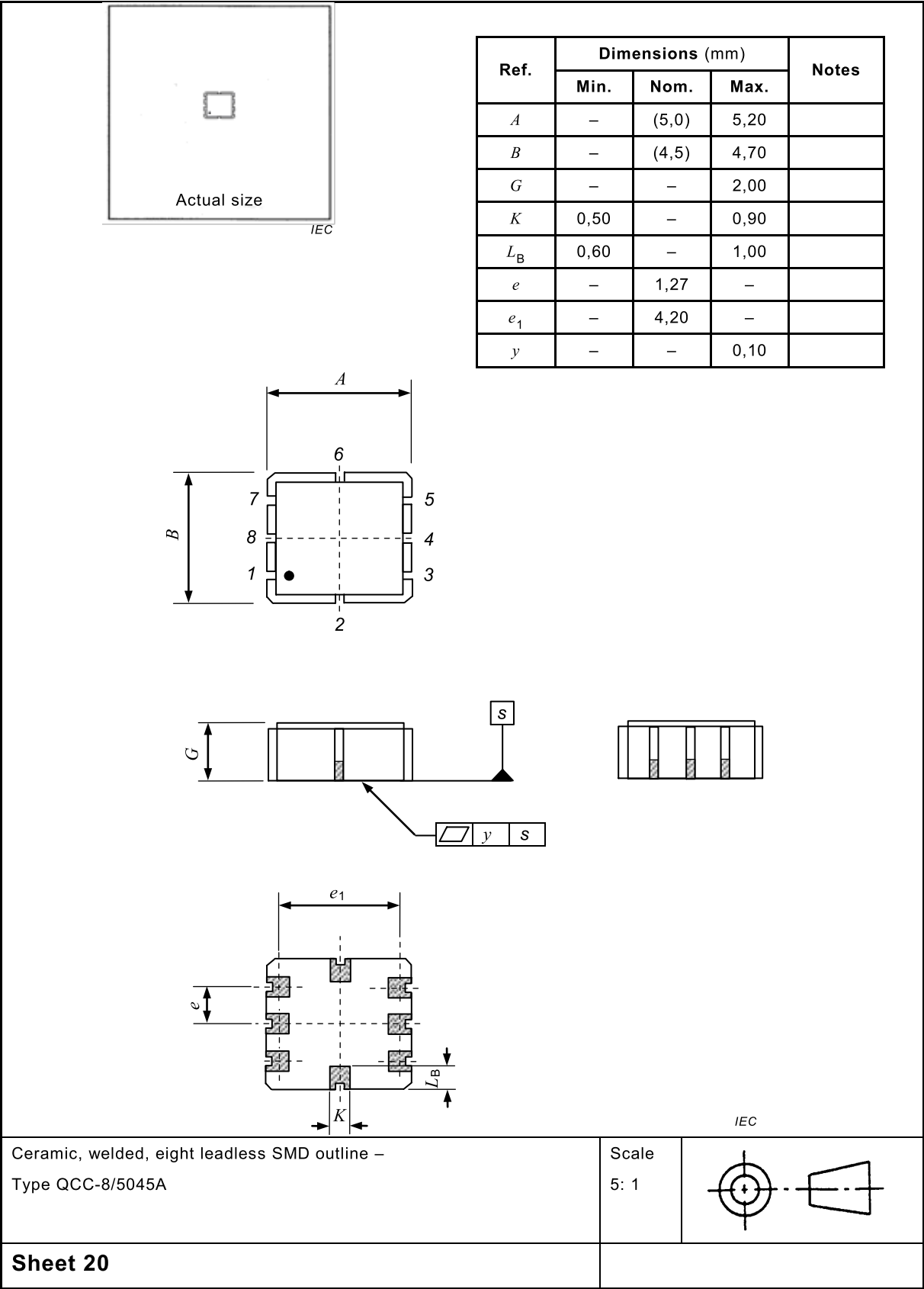
Scale
5: 1



Sheet 19

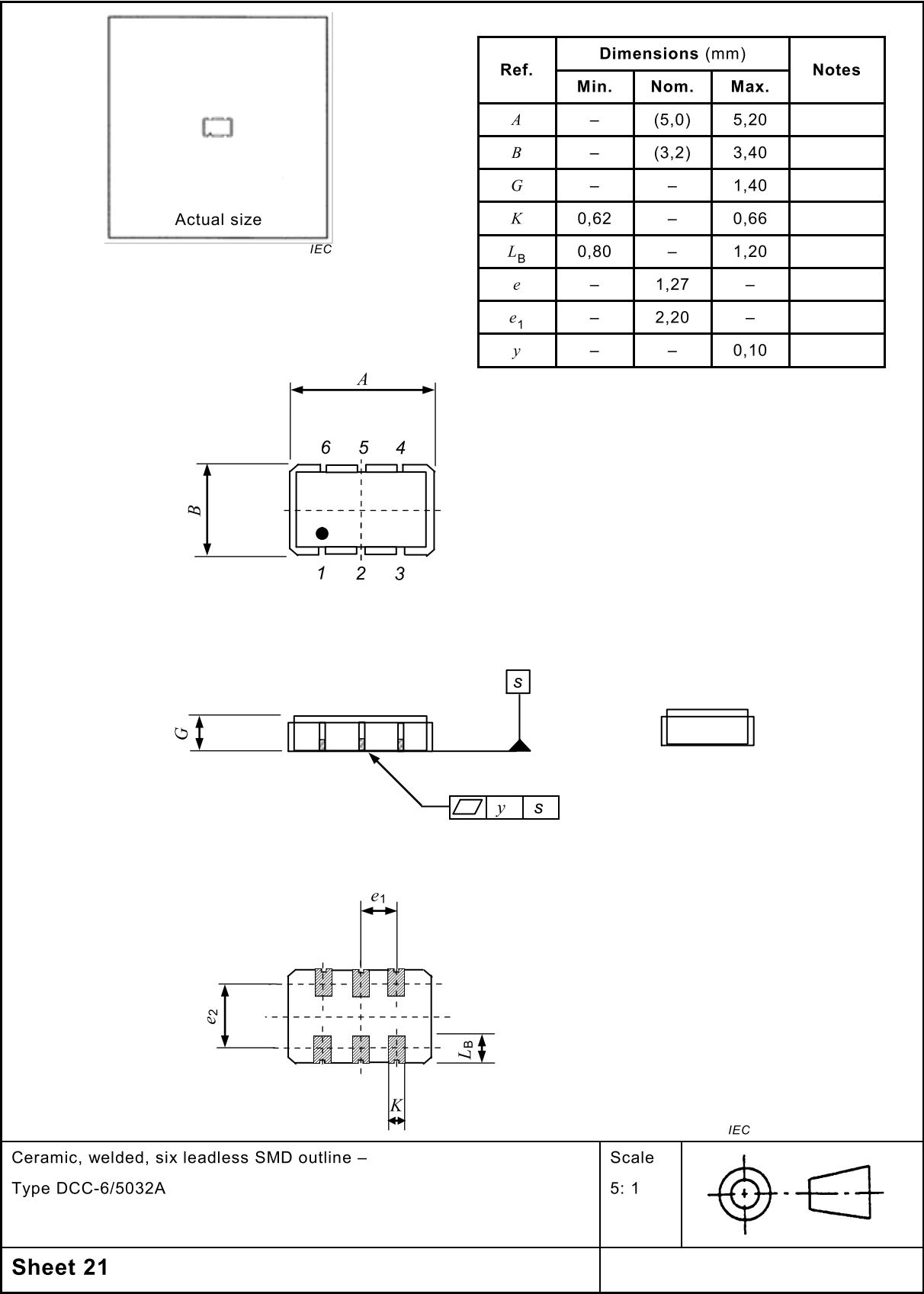
Terminal land connections of Type QCC-12/5045A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1				Ground
2				To be specified
3				Ground
4				Input/Output /Ground
5				Output/Input
6				Output/Input /Ground
7				Ground
8				To be specified
9				Ground
10				Output/Input /Ground
11				Output/Input
12				Output/Input /Ground



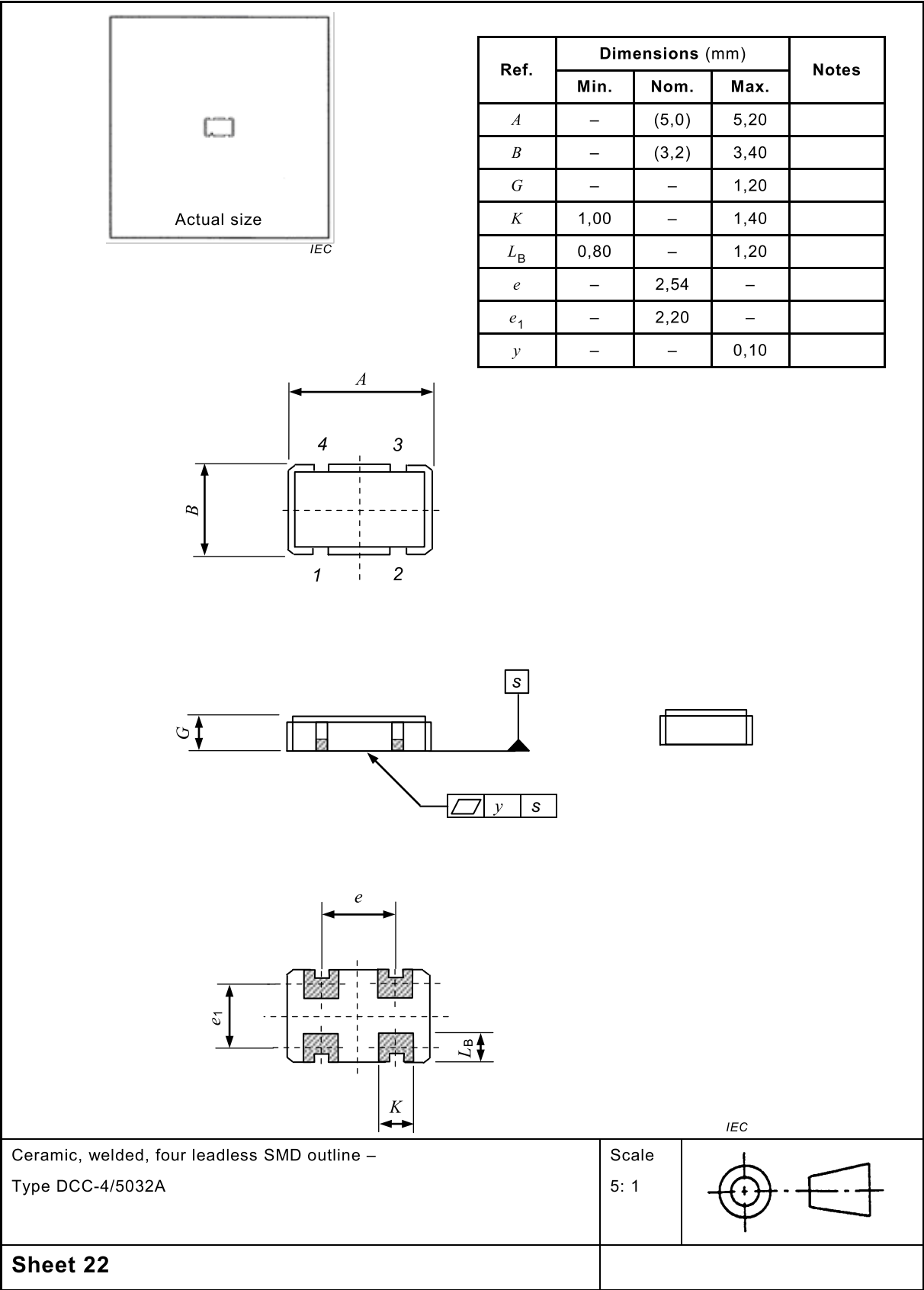
Terminal land connections of Type QCC-8/5045A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Control voltage	Ground	Ground
2		Option	Ground	Ground
3		Ground	Ground	Ground
4		Option	Input/Output	Input/Output
5		Output	Ground	Ground
6		Option	Ground	Ground
7		DC supply	Ground	Ground
8		Option	Output/Input	Output/Input



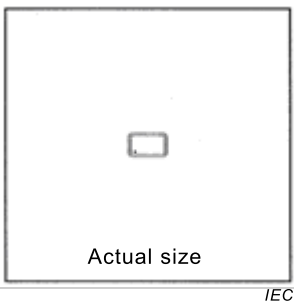
Terminal land connections of Type DCC-6/5032A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Output Enable		
2		NC		
3		Ground		
4		Output		
5		Output		
6		DC supply		

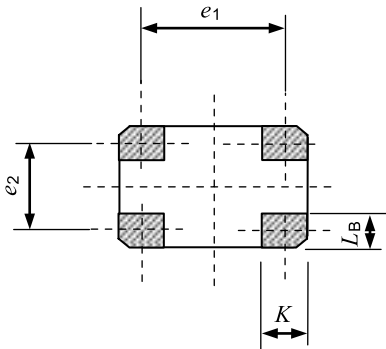
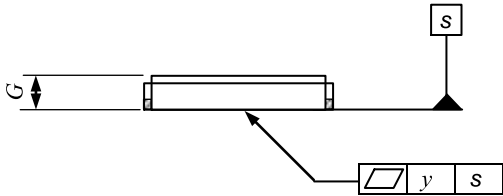
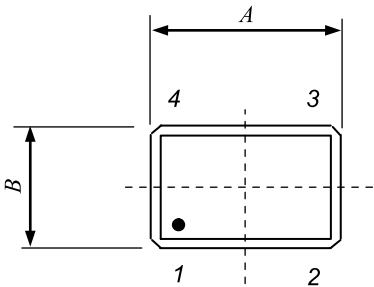


Terminal land connections of Type DCC-4/5032A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Control voltage		
2		Ground		
3		Output		
4		DC supply		



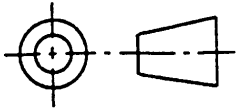
Ref.	Dimensions (mm)			Notes
	M□n.	□om.	Max.	
<i>A</i>	–	(5,0)	5,20	
<i>B</i>	–	(3,2)	3,40	
<i>G</i>	–	–	0,90	
<i>K</i>	1,00	–	1,40	
<i>L_B</i>	0,70	–	1,10	
<i>e₁</i>	–	3,80	–	
<i>e₂</i>	–	2,30	–	
<i>y</i>	–	–	0,10	



IEC

Ceramic, welded, four leadless SMD outline –
Type DCC-4/5032C

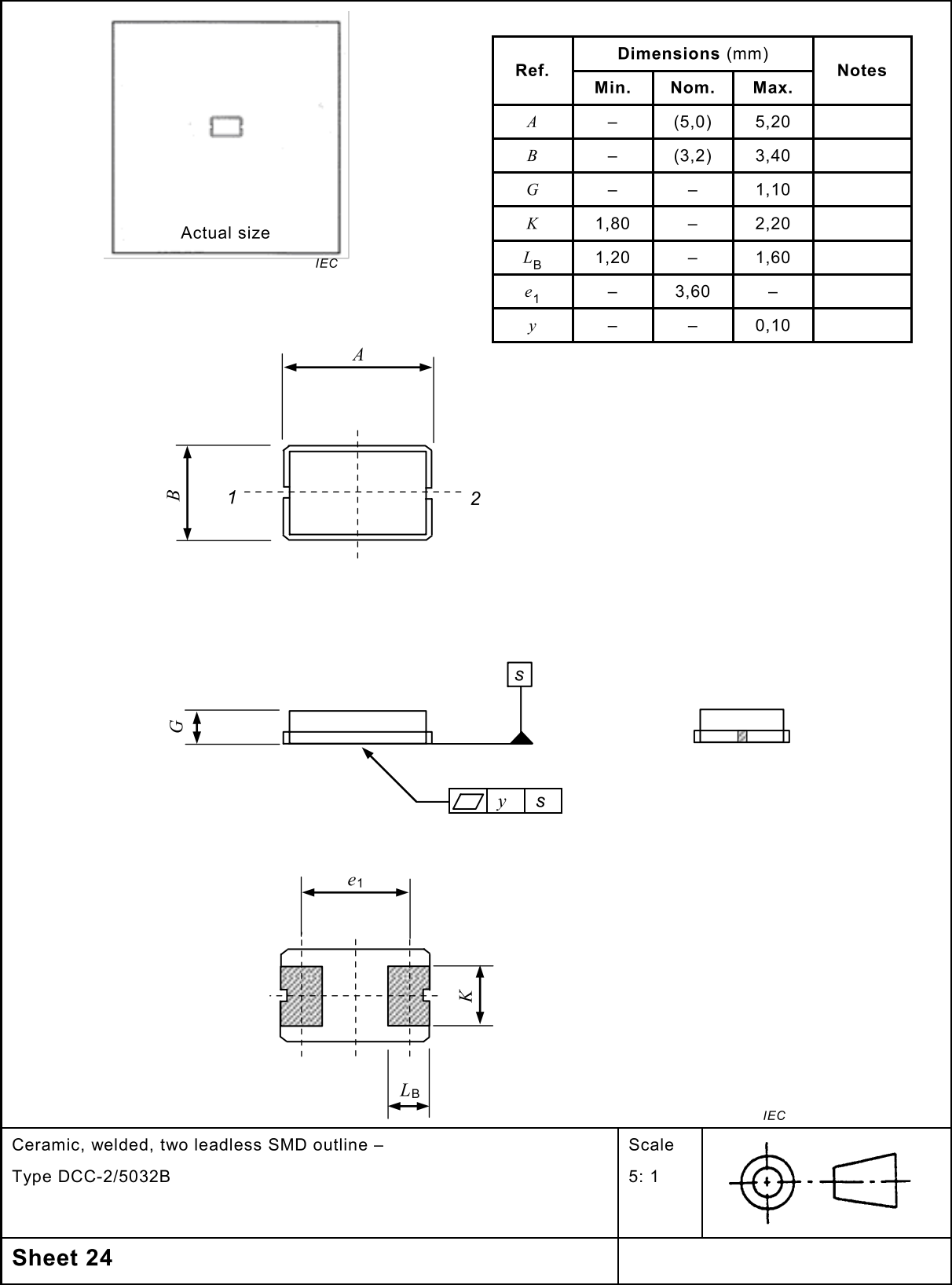
Scale
5: 1



Sheet 23

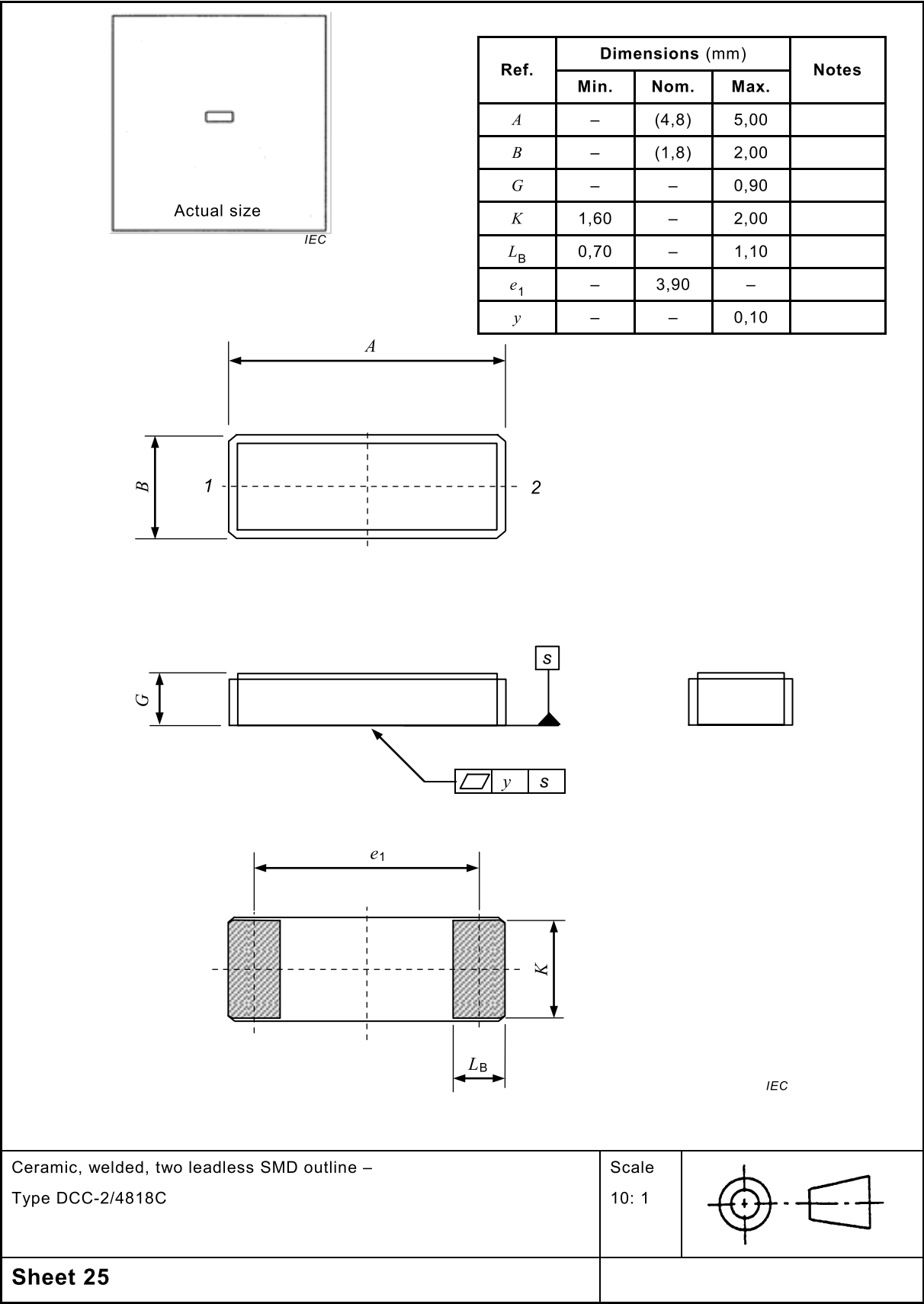
Terminal land connections of Type DCC-4/5032C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1	Option		
2	Ground	Ground		
3	Terminal 2	Output		
4	Ground	DC supply		



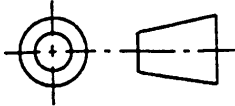
Terminal land connections of Type DCC-2/5032B

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			



Ceramic, welded, two leadless SMD outline –
Type DCC-2/4818C

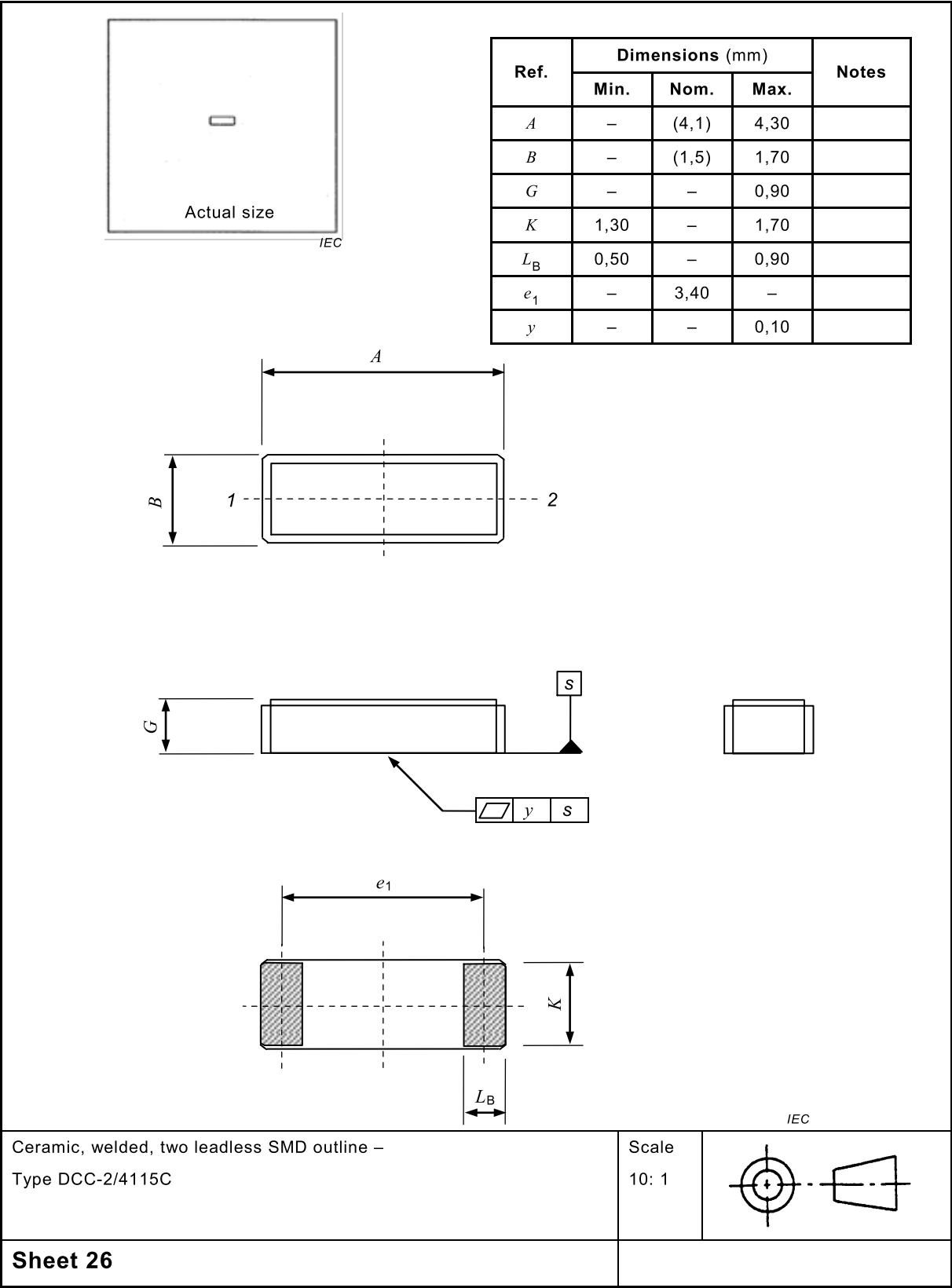
Scale
10: 1



Sheet 25

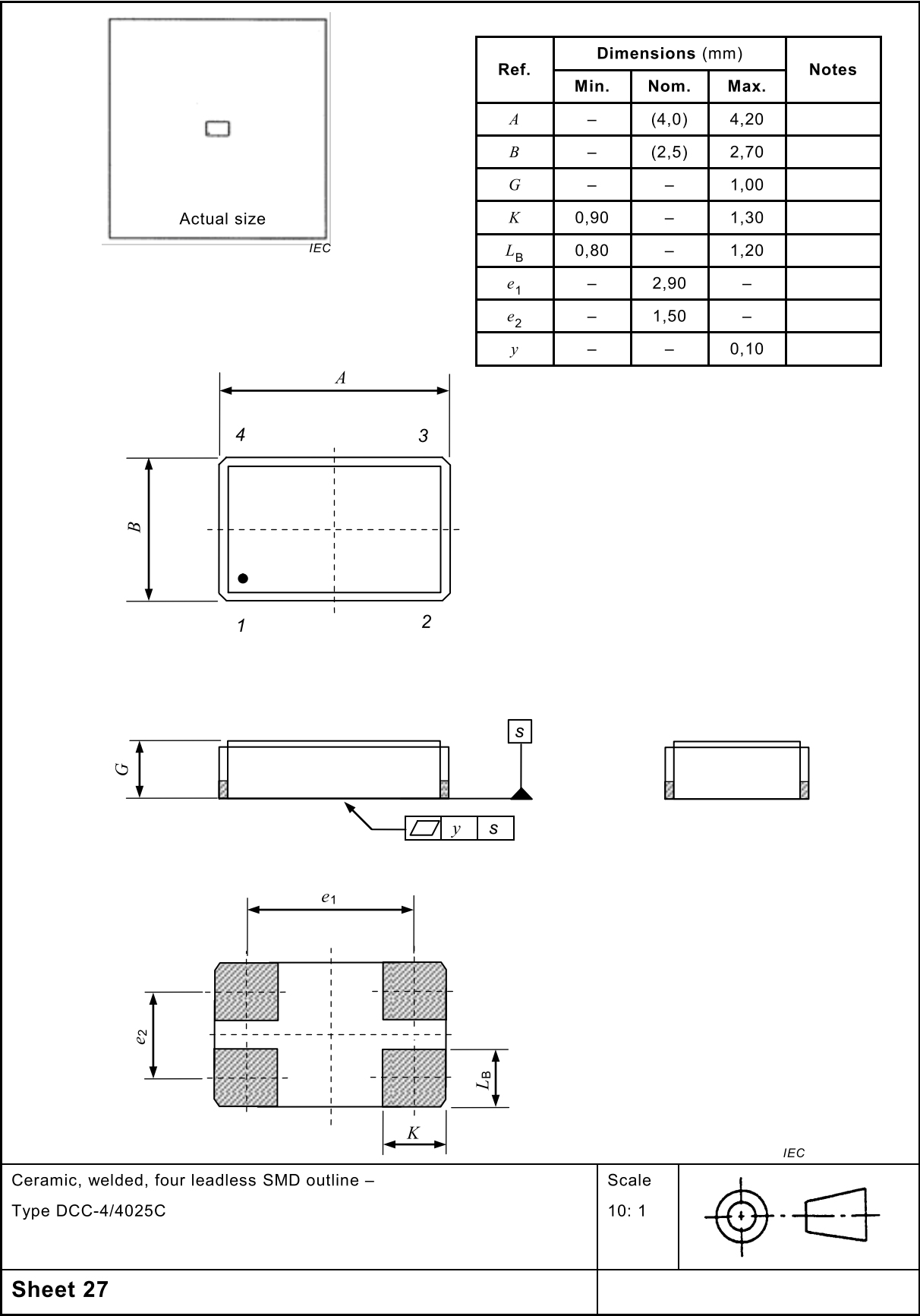
Terminal land connections of Type DCC-2/4818C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			



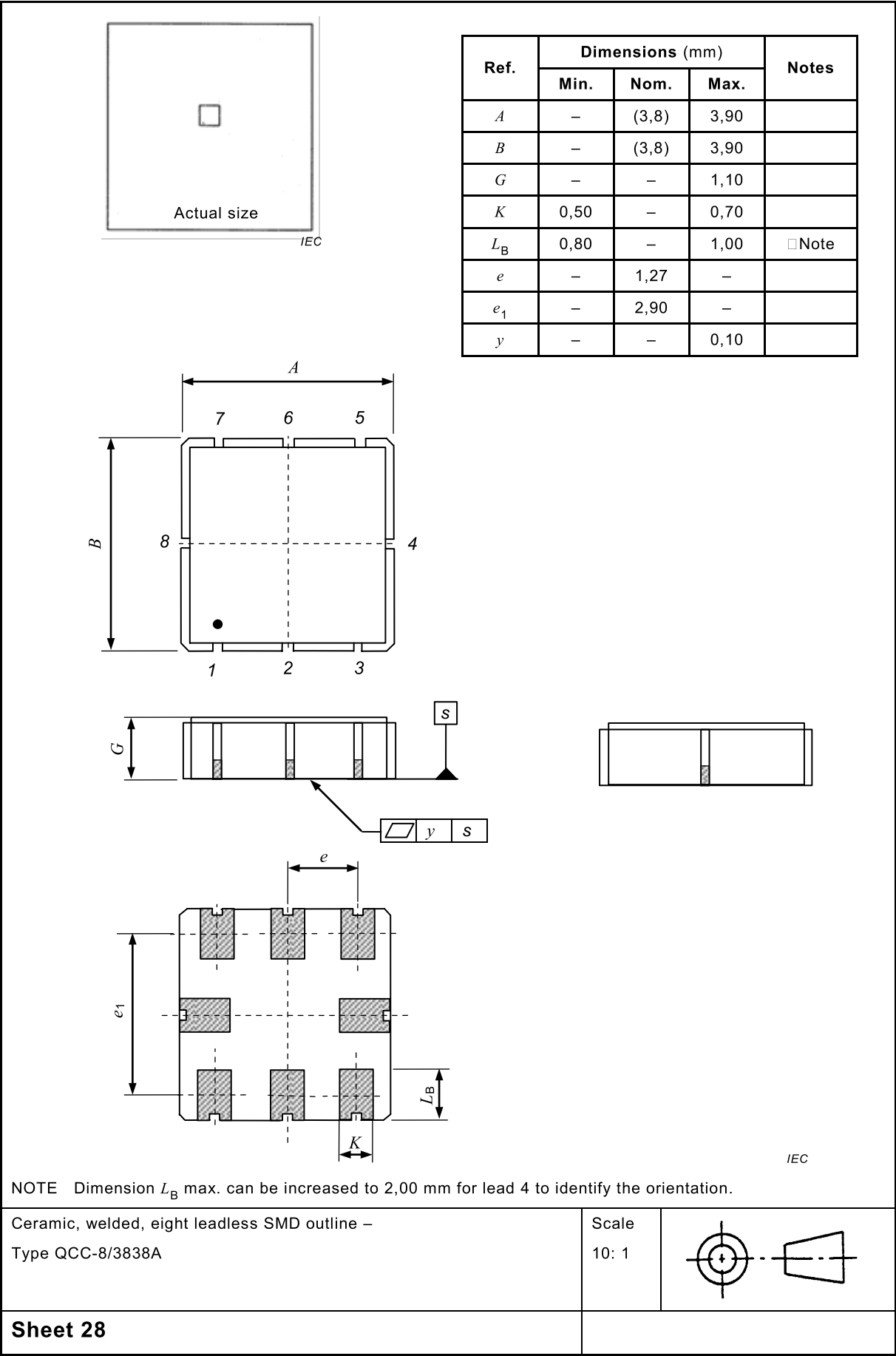
Terminal land connections of Type DCC-2/4115C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			



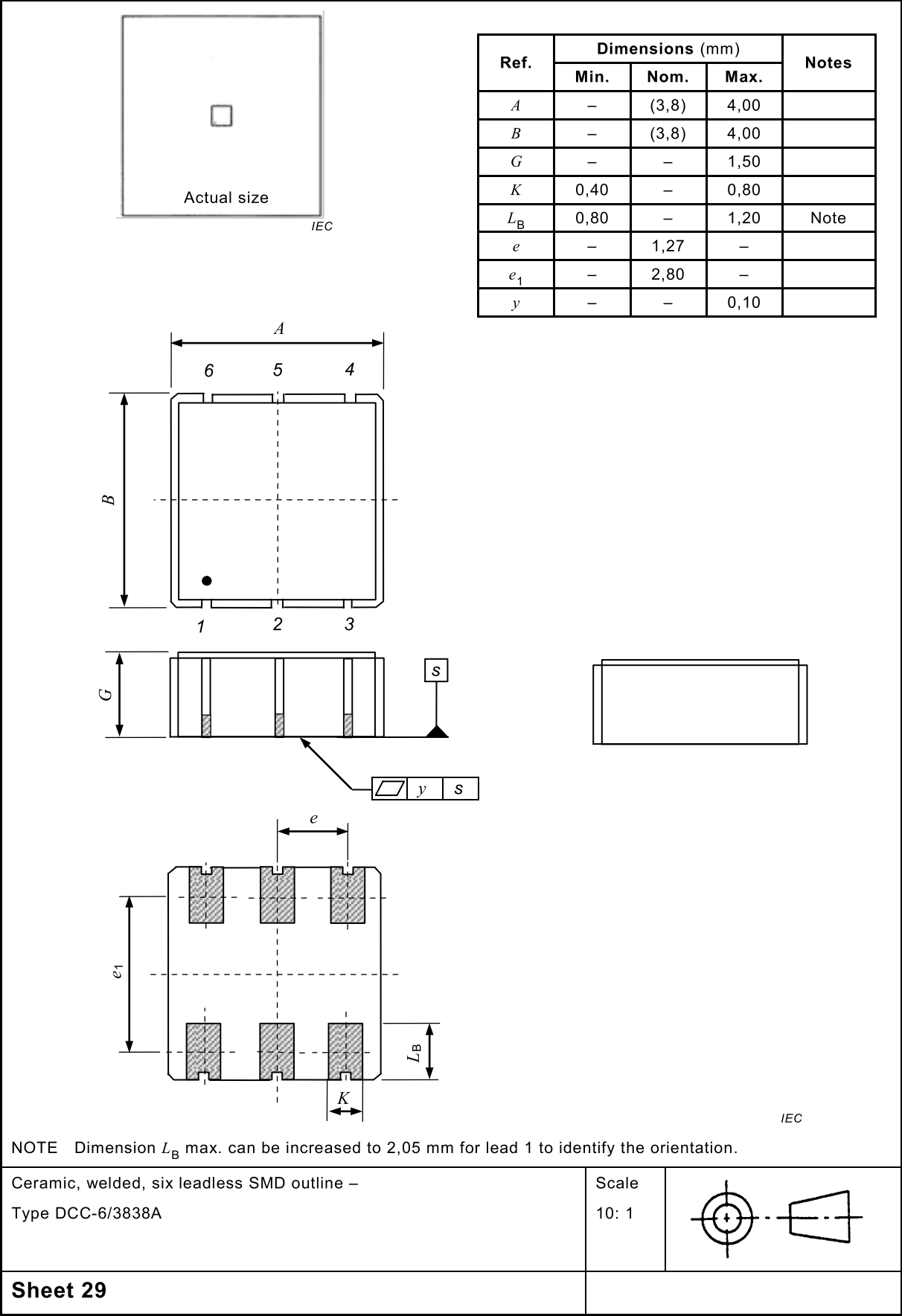
Terminal land connections of Type DCC-4/4025C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1	Control Voltage		
2	Ground	Ground		
3	Terminal 2	Output		
4	Ground	DC supply		



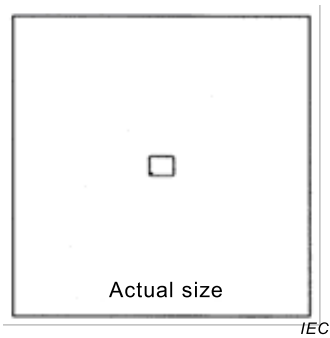
Terminal land connections of Type QCC-8/3838A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1			Input/Output	Input/Output
2			Ground	Ground
3			Ground	Option
4			Ground	Ground
5			Output/Input	Option
6			Ground	Ground
7			Ground	Output/Input
8			Ground	Ground

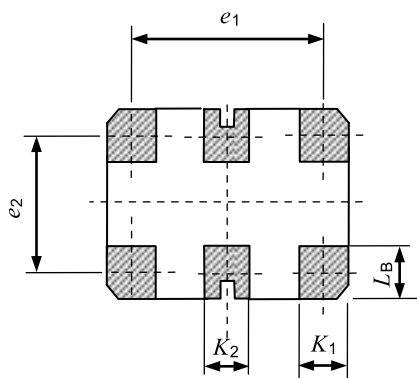
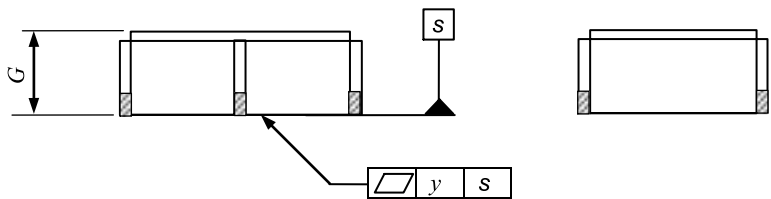
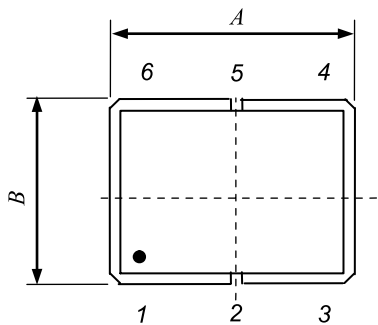


Terminal land connections of Type DCC-6/3838A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1				Ground
2				Input/Output
3				Ground
4				Ground
5				Output/Input
6				Ground



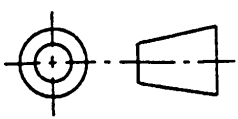
Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
A	–	(3,2)	3,40	
B	–	(2,5)	2,70	
G	–	–	1,10	
K_1	0,45	–	0,85	
K_2	0,40	–	0,80	
L_B	0,50	–	0,90	
e_1	–	2,55	–	
e_2	–	1,80	–	
y	–	–	0,10	



IEC

Ceramic, welded, six leadless SMD outline –
Type DCC-6/3225A

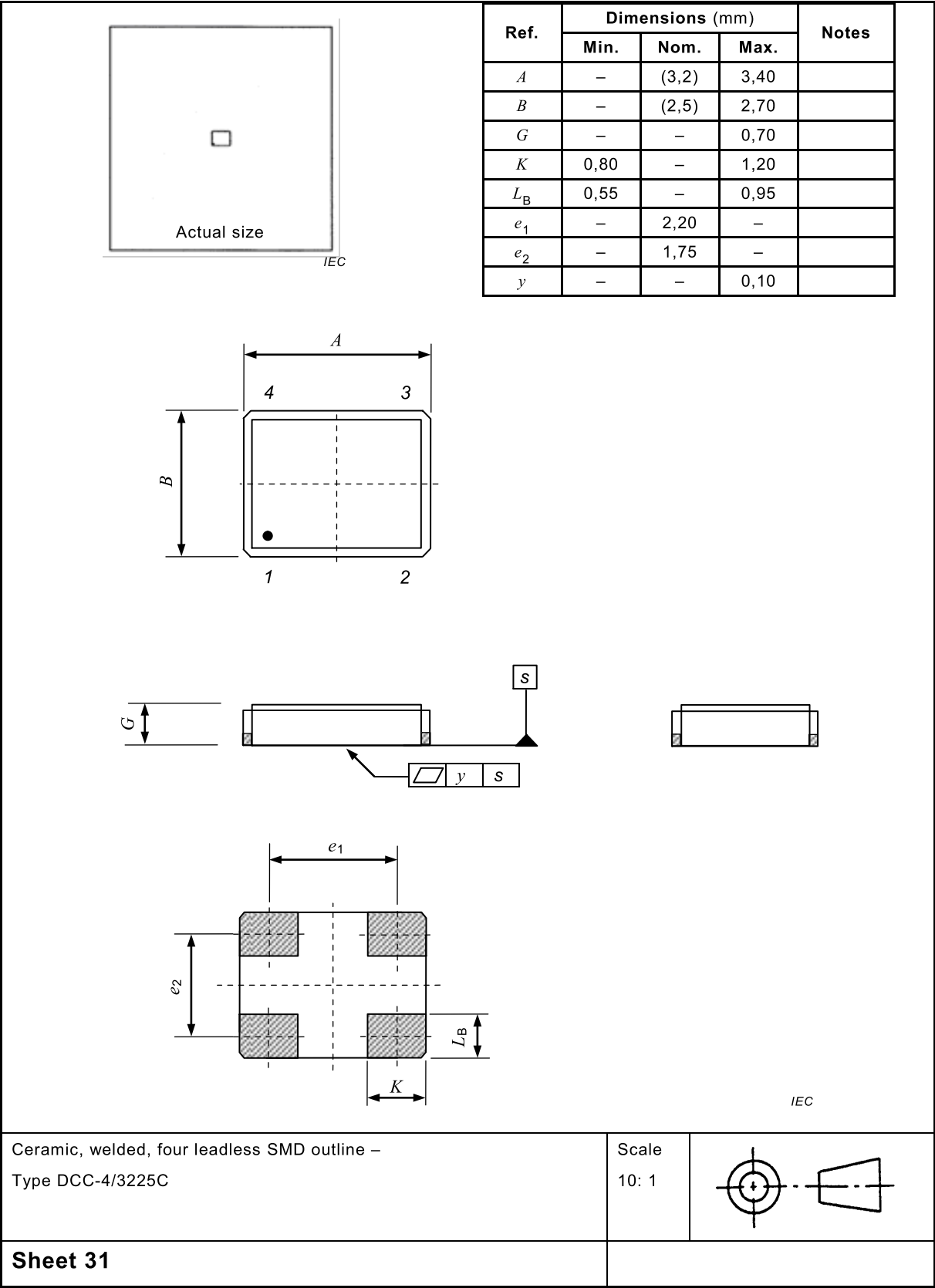
Scale
10: 1



Sheet 30

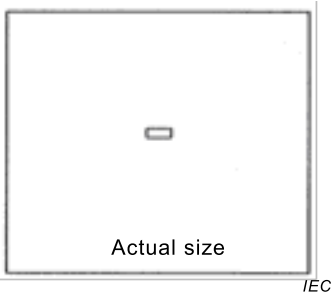
Terminal land connections of Type DCC-6/3225A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Output Enable		
2		NC		
3		Ground		
4		Output		
5		Output		
6		DC supply		

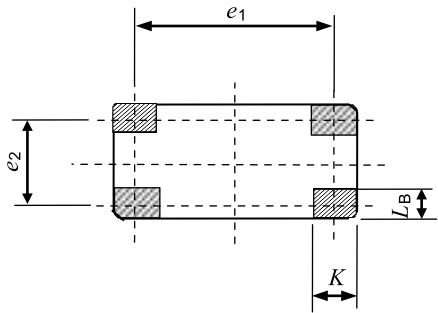
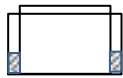
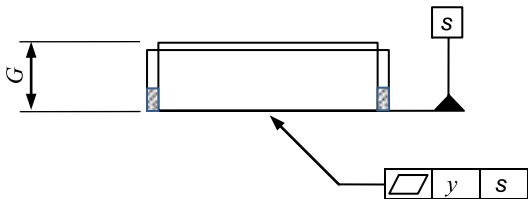
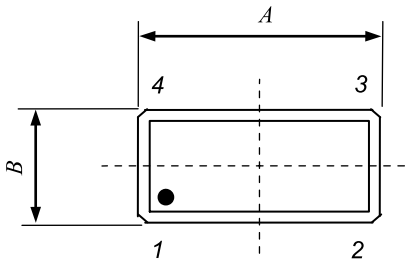


Terminal land connections of Type DCC-4/3225C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1	Control voltage		
2	Option	Ground		
3	Terminal 2	Output		
4	Option	DC supply		



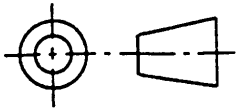
Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
<i>A</i>	–	(3,20)	3,40	
<i>B</i>	–	(1,50)	1,70	
<i>G</i>	–	–	1,00	
<i>K</i>	0,40	–	0,80	
<i>L_B</i>	0,20	–	0,60	
<i>e₁</i>	–	2,80	–	
<i>e₂</i>	–	0,90		
<i>y</i>	–	–	0,10	



IEC

Ceramic, welded, four leadless SMD outline –
Type DCC-4/3215C

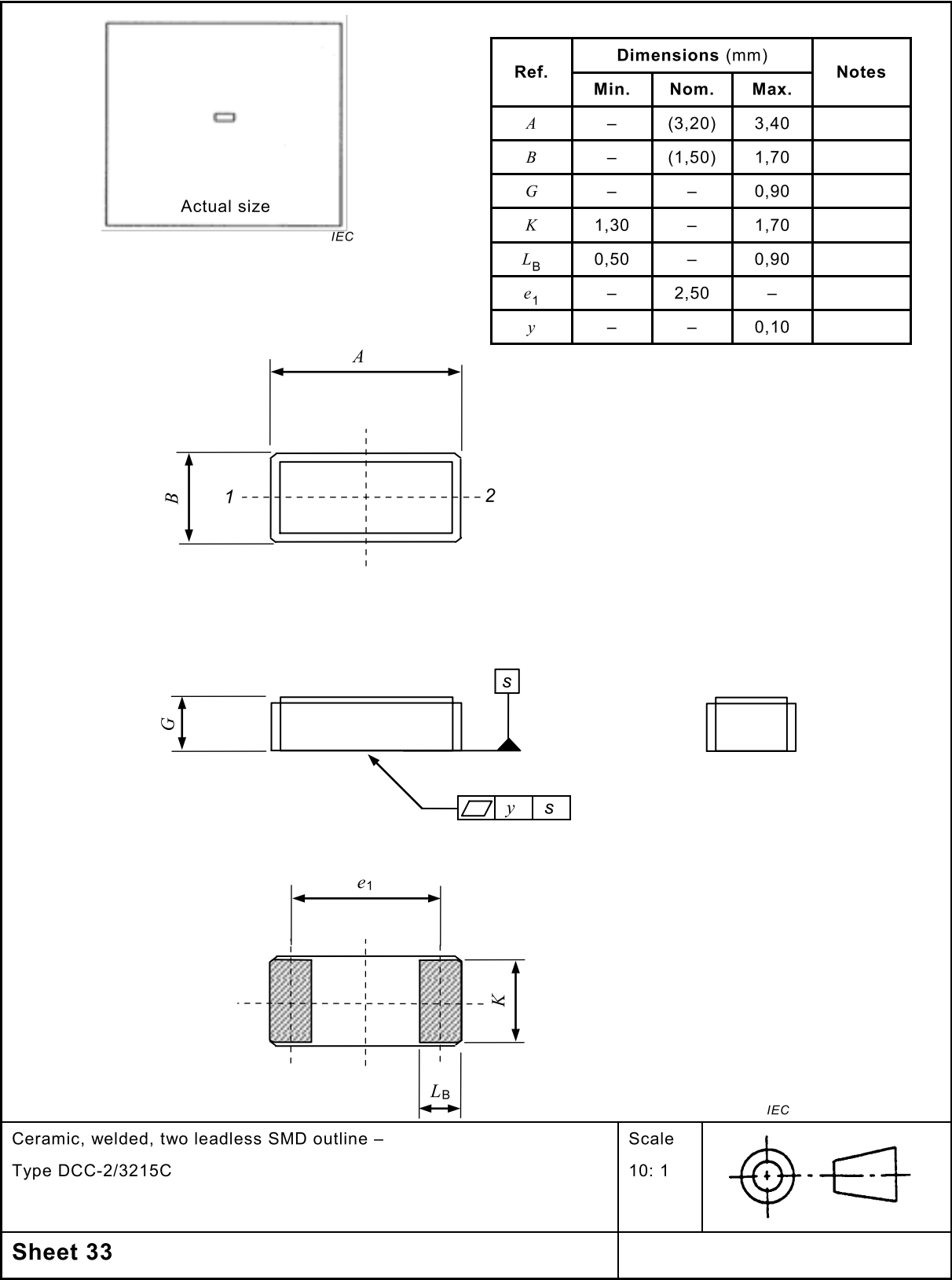
Scale
10: 1



Sheet 32

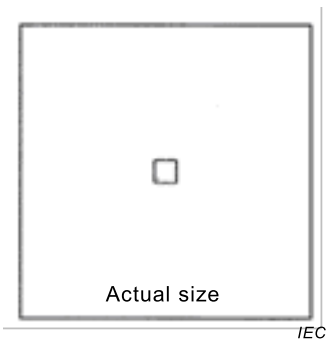
Terminal land connections of Type DCC-4/3215C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1	Control voltage		
2	Ground	Ground		
3	Ground	Output		
4	Terminal 2	DC supply		

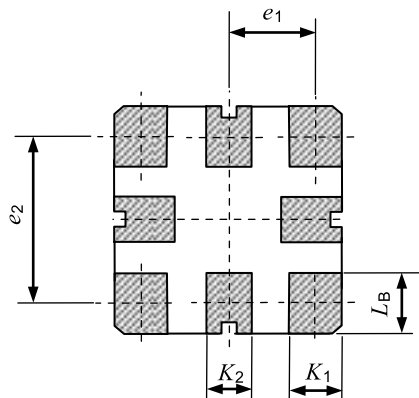
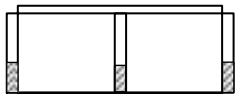
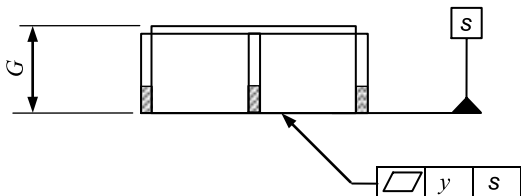
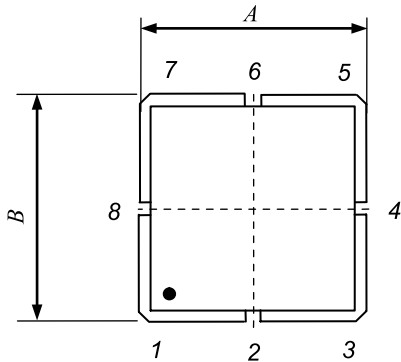


Terminal land connections of Type DCC-2/3215C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			



Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
<i>A</i>	–	(3,0)	3,20	
<i>B</i>	–	(3,0)	3,20	
<i>G</i>	–	–	1,15	
<i>K</i> ₁	0,50	–	0,90	
<i>K</i> ₂	0,40	–	0,80	
<i>L</i> _B	0,70	–	0,90	Note
<i>e</i> ₁	–	1,15	–	
<i>e</i> ₂	–	2,20	–	
<i>y</i>	–	–	0,10	

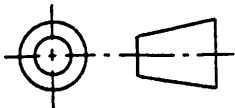


IEC

NOTE Dimension *L_B* max. can be increased to 1,60 mm for lead 4 to identify the orientation.

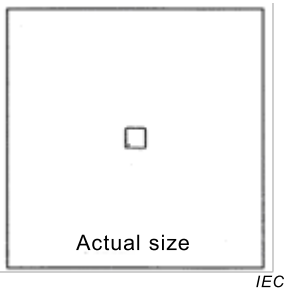
Ceramic, welded, eight leadless SMD outline –
Type QCC-8/3030B

Scale
10: 1

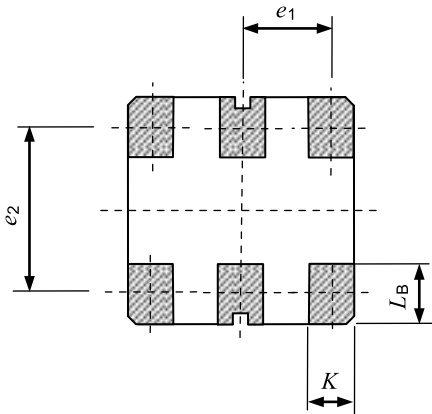
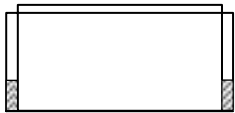
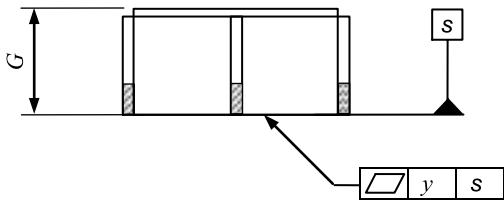
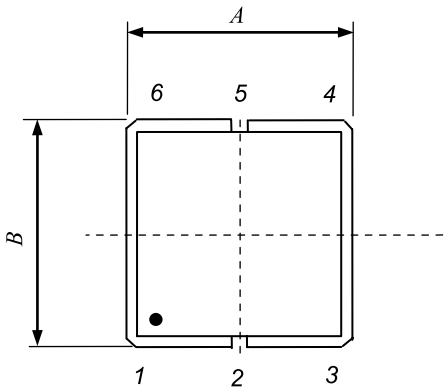


Terminal land connections of Type QCC-8/3030B

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1			Input/Output	Input/Output
2			Ground	Ground
3			Ground	Option
4			Ground	Ground
5			Output/Input	Option
6			Ground	Ground
7			Ground	Output/Input
8			Ground	Ground



Ref. □	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
<i>A</i>	–	(3,0)	3,20	
<i>B</i>	–	(3,0)	3,20	
<i>G</i>	–	–	1,40	
<i>K</i>	0,40	–	0,80	
<i>L_B</i>	0,60	–	1,00	Note
<i>e₁</i>	–	1,20	–	
<i>e₂</i>	–	2,20	–	
<i>y</i>	–	–	0,10	

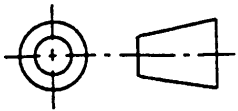


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NOTE Dimension L_B max. can be increased to 1,60 mm for lead 6 to identify the orientation.

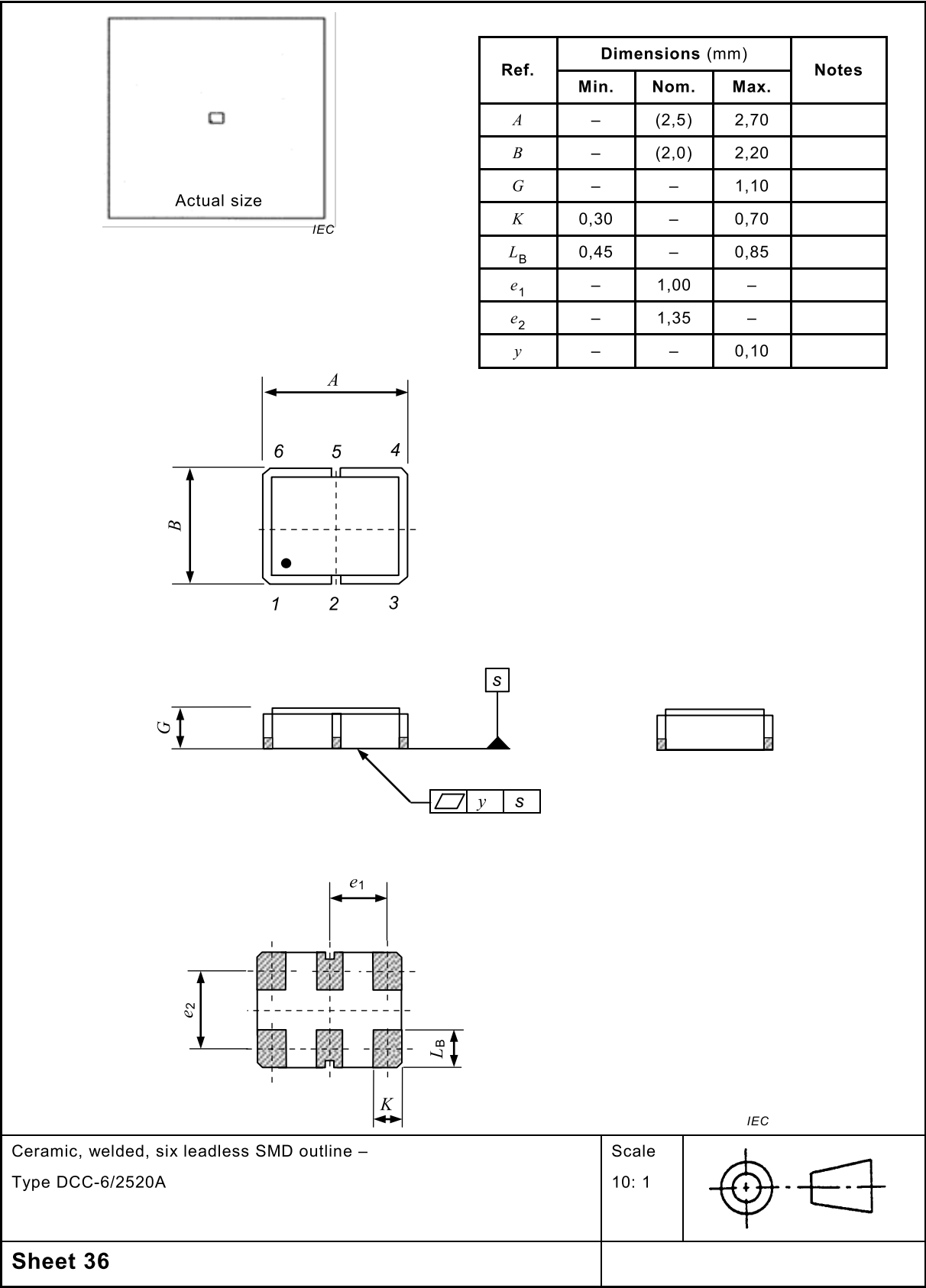
Ceramic, welded, six leadless SMD outline –
Type DCC-6/3030A

Scale
10: 1



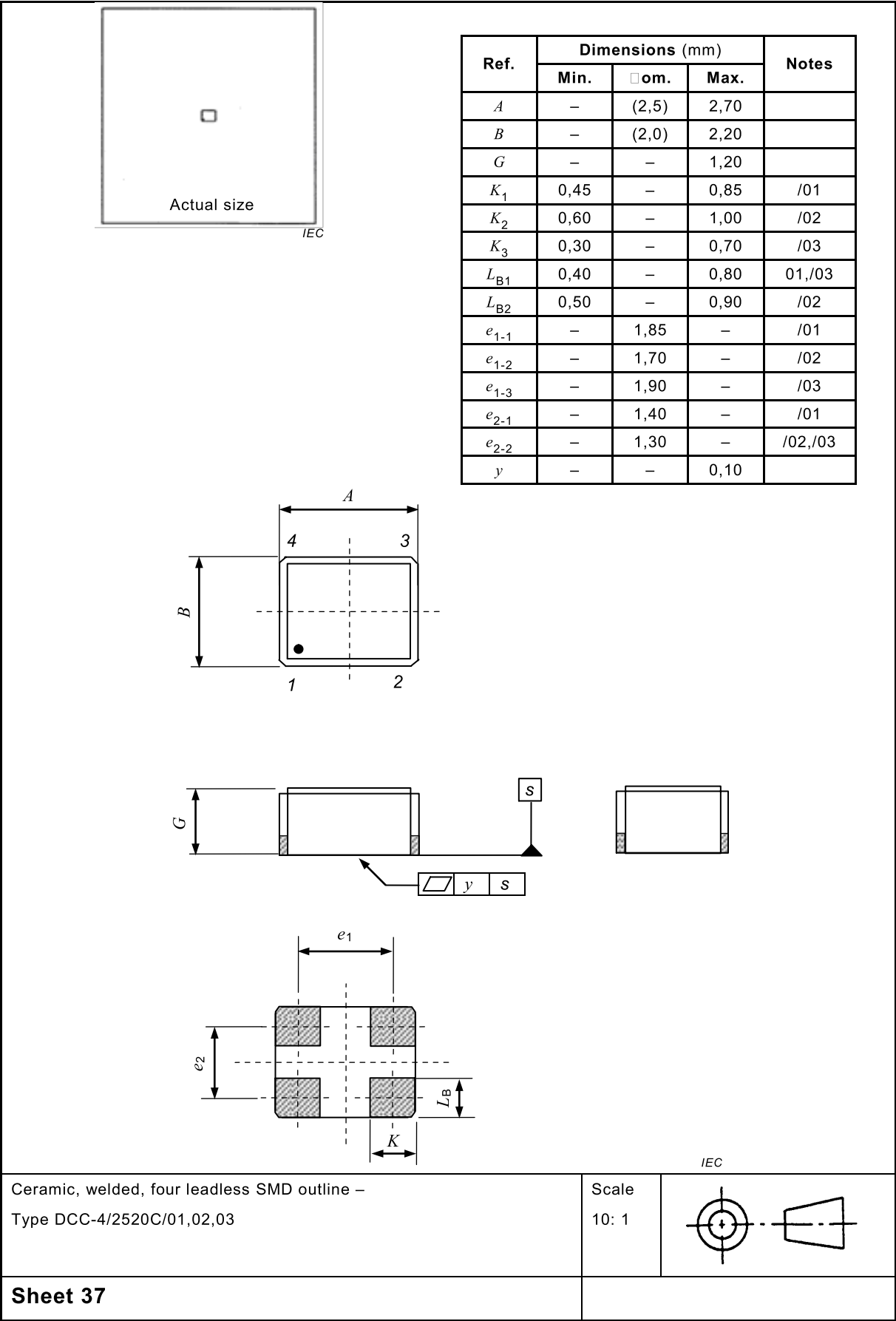
Terminal land connections of Type DCC-6/3030A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1				Ground
2				Input/Output
3				Ground
4				Ground
5				Output/Input
6				Ground



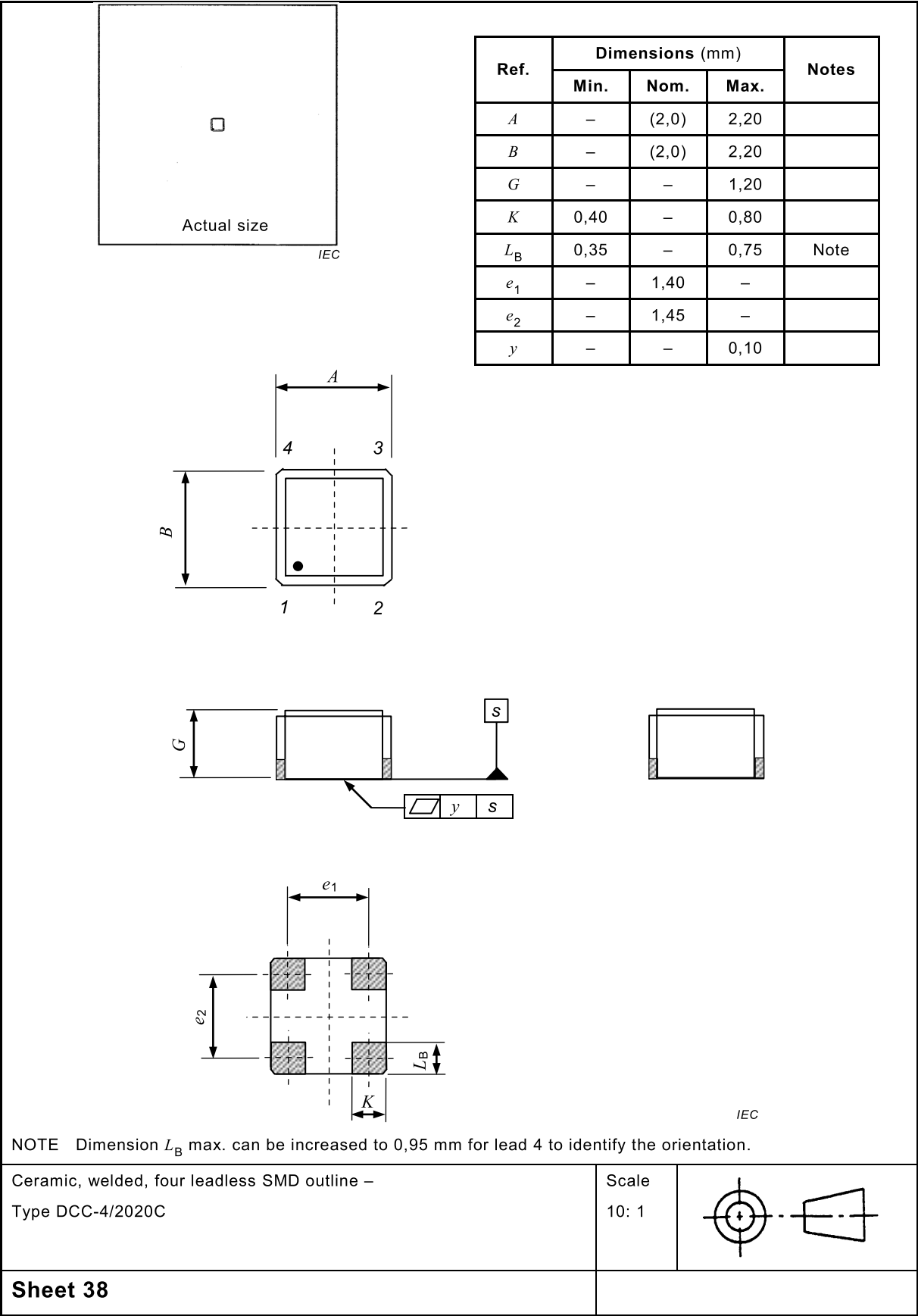
Terminal land connections of Type DCC-6/2520A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Option	Option	
2		Option	Ground	
3		Ground	Output/Input	
4		Output	Option	
5		Option	Ground	
6		DC supply	Input/Output	



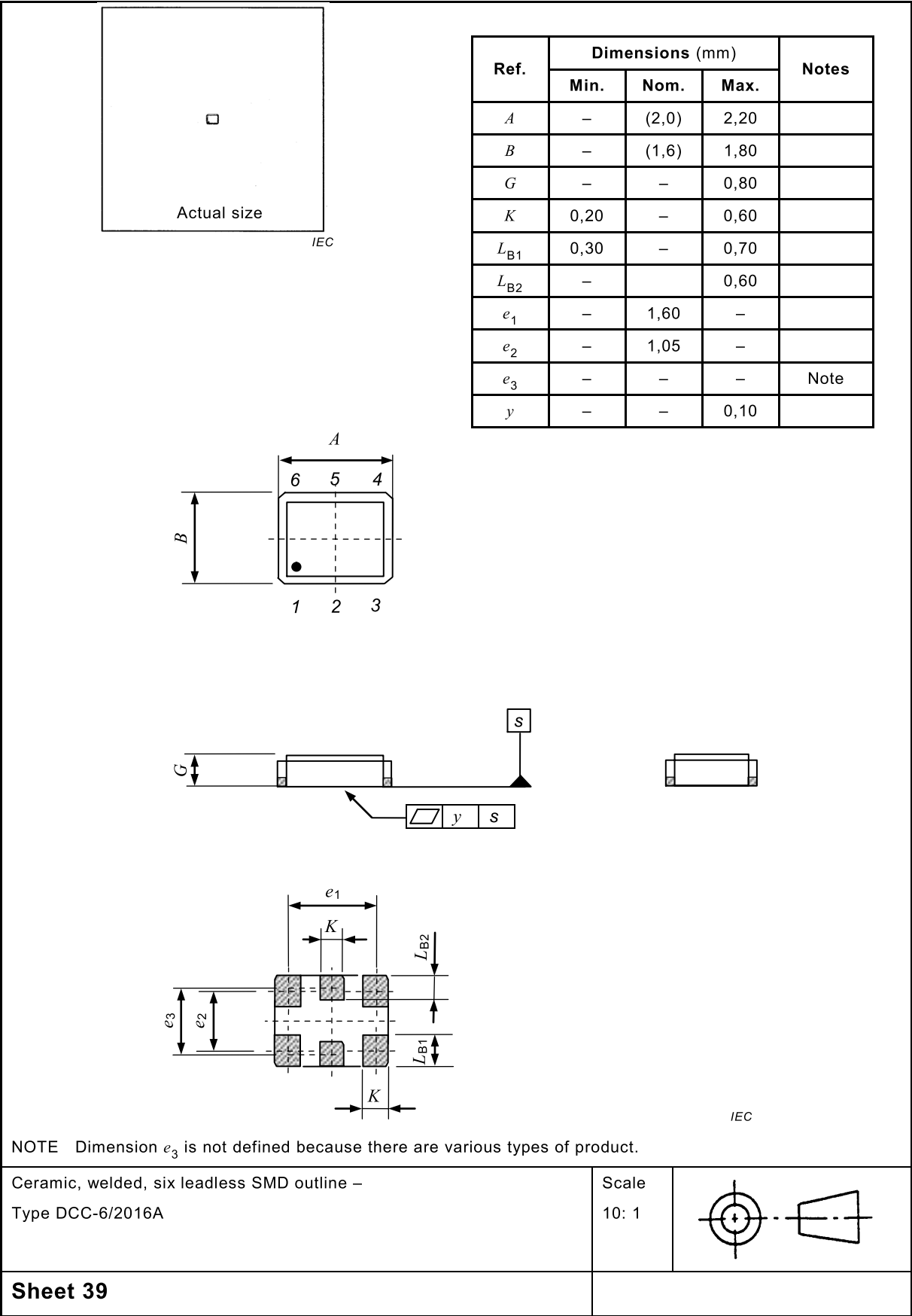
Terminal land connections of Type DCC-4/2520C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1	Option		
2	Ground/NC	Ground		
3	Terminal 2	Output		
4	Option	DC supply		



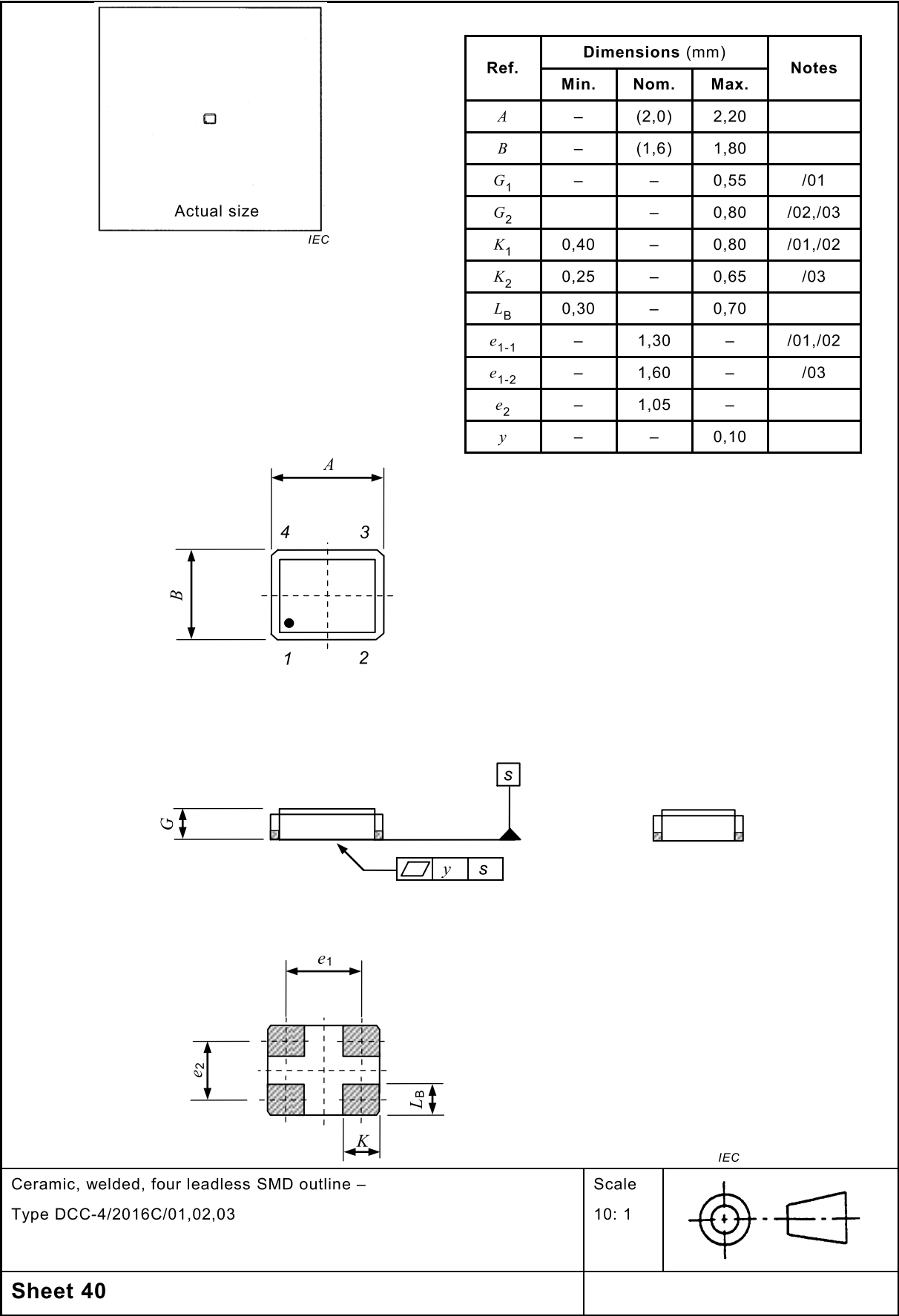
Terminal land connections of Type DCC-4/2020C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1		Ground	Ground
2	Ground		Output/Input	Output/Input
3	Terminal 2		Ground	Input/Output
4	Ground		Input/ Output	Ground



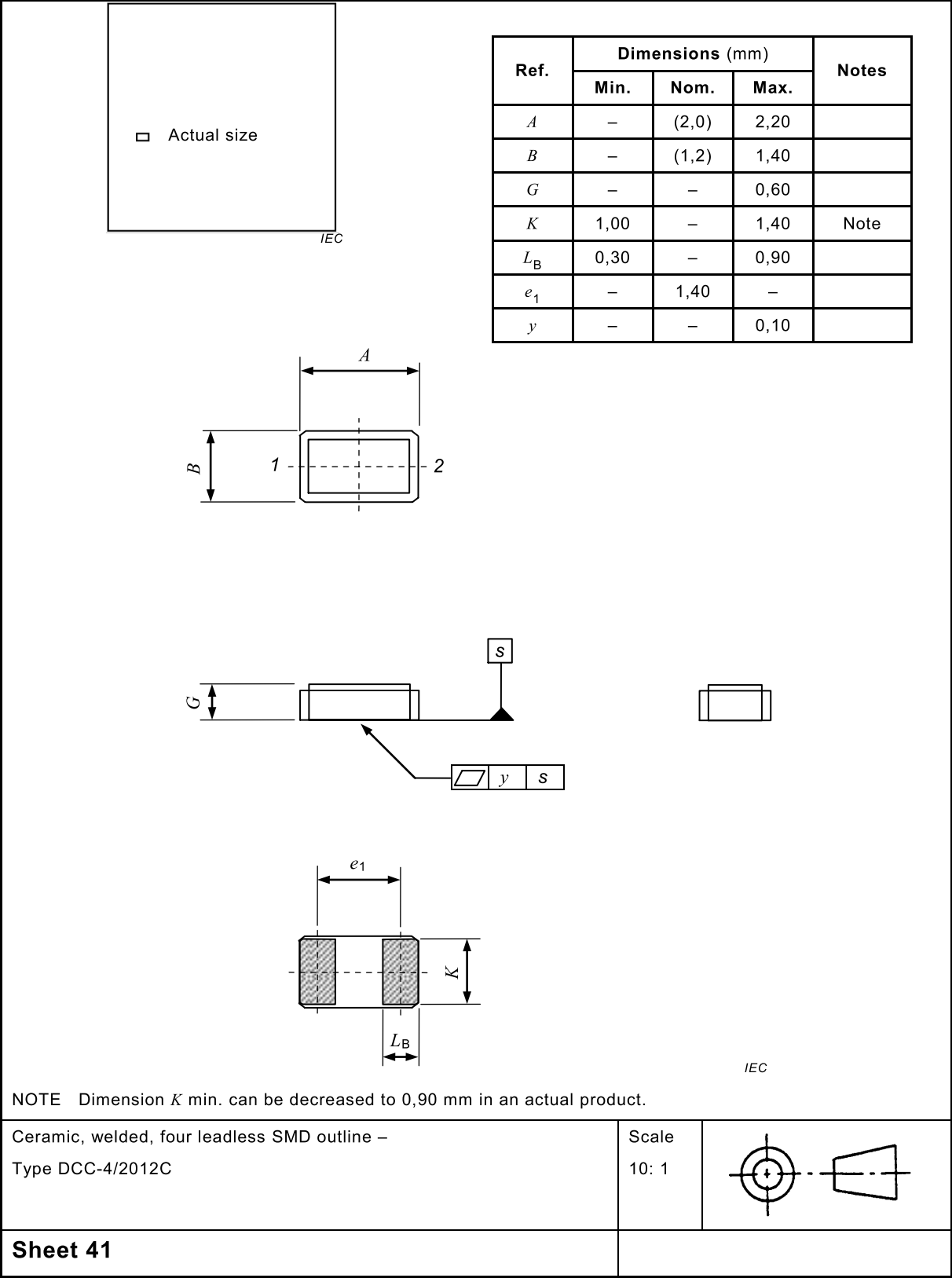
Terminal land connections of Type DCC-6/2016A

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1		Option		
2		Option		
3		Ground		
4		Output		
5		Option		
6		DC supply		



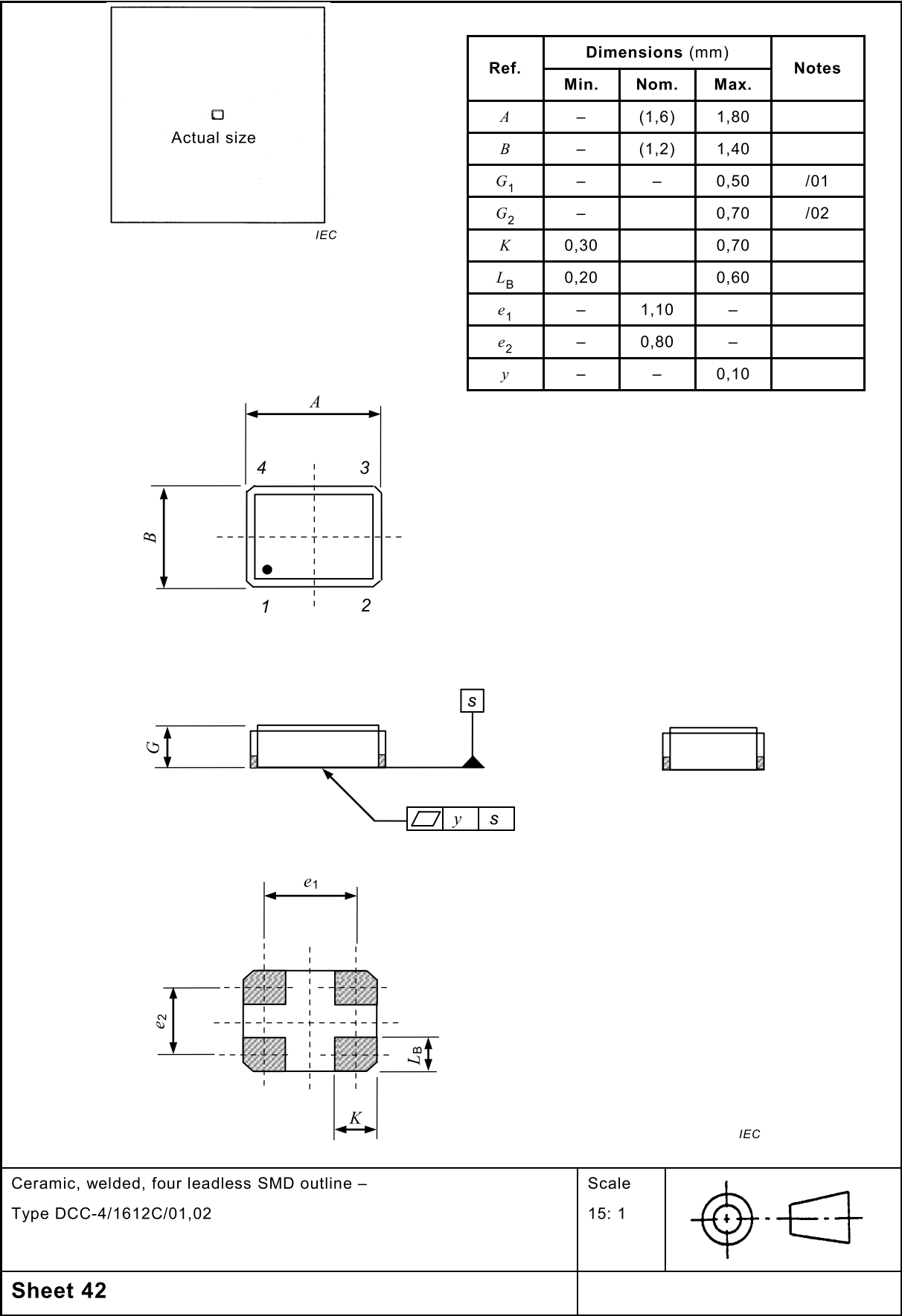
Terminal land connections of Type DCC-4/2016C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Ground			
3	Terminal 2			
4	Ground			



Terminal land connections of Type DCC-2/2012C

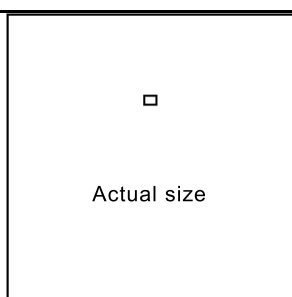
No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			



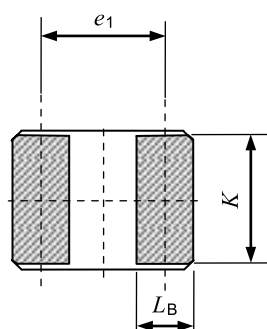
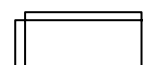
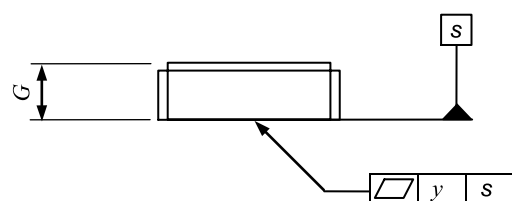
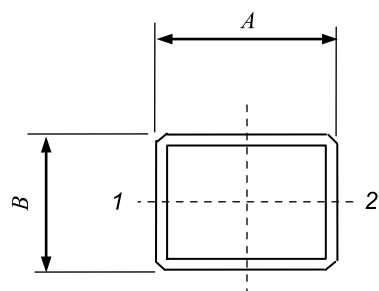
Terminal lead connections of Type DCC-4/1612C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW device
1	Terminal 1	Option		
2	Ground	Ground		
3	Terminal 2	Output		
4	Ground	DC supply		

Ref.	Dimensions (mm)			Notes
	Min.	Nom.	Max.	
A	—	(1,6)	1,80	
B	—	(1,2)	1,40	
G	—	—	0,50	
K	1,00	—	1,40	
L_B	0,30	—	0,70	
e_1	—	1,10	—	
y	—	—	0,10	



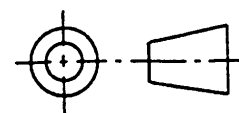
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Ceramic, welded, two leadless SMD outline –
Type DCC-2/1612C

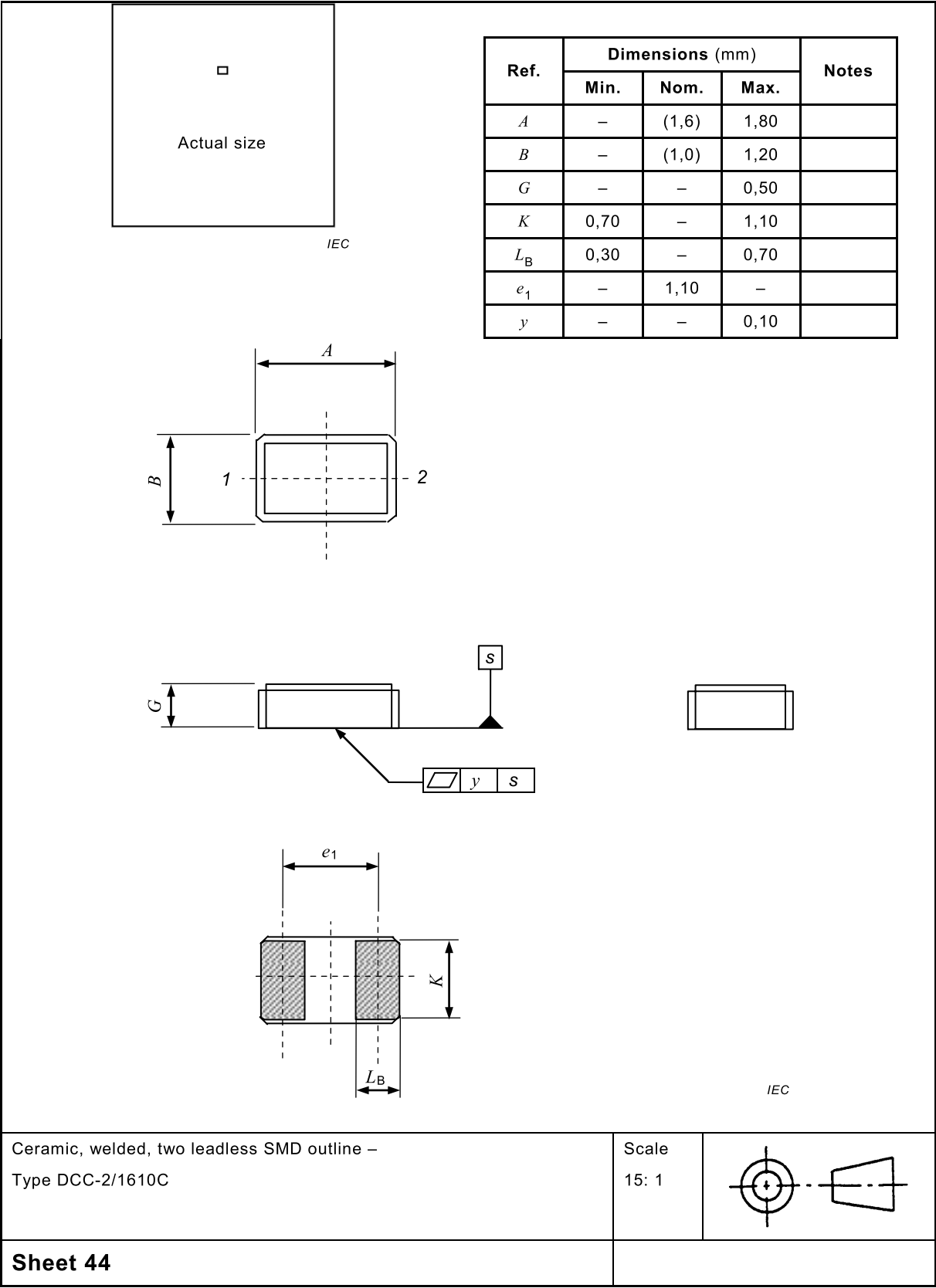
Scale
15: 1



Sheet 43

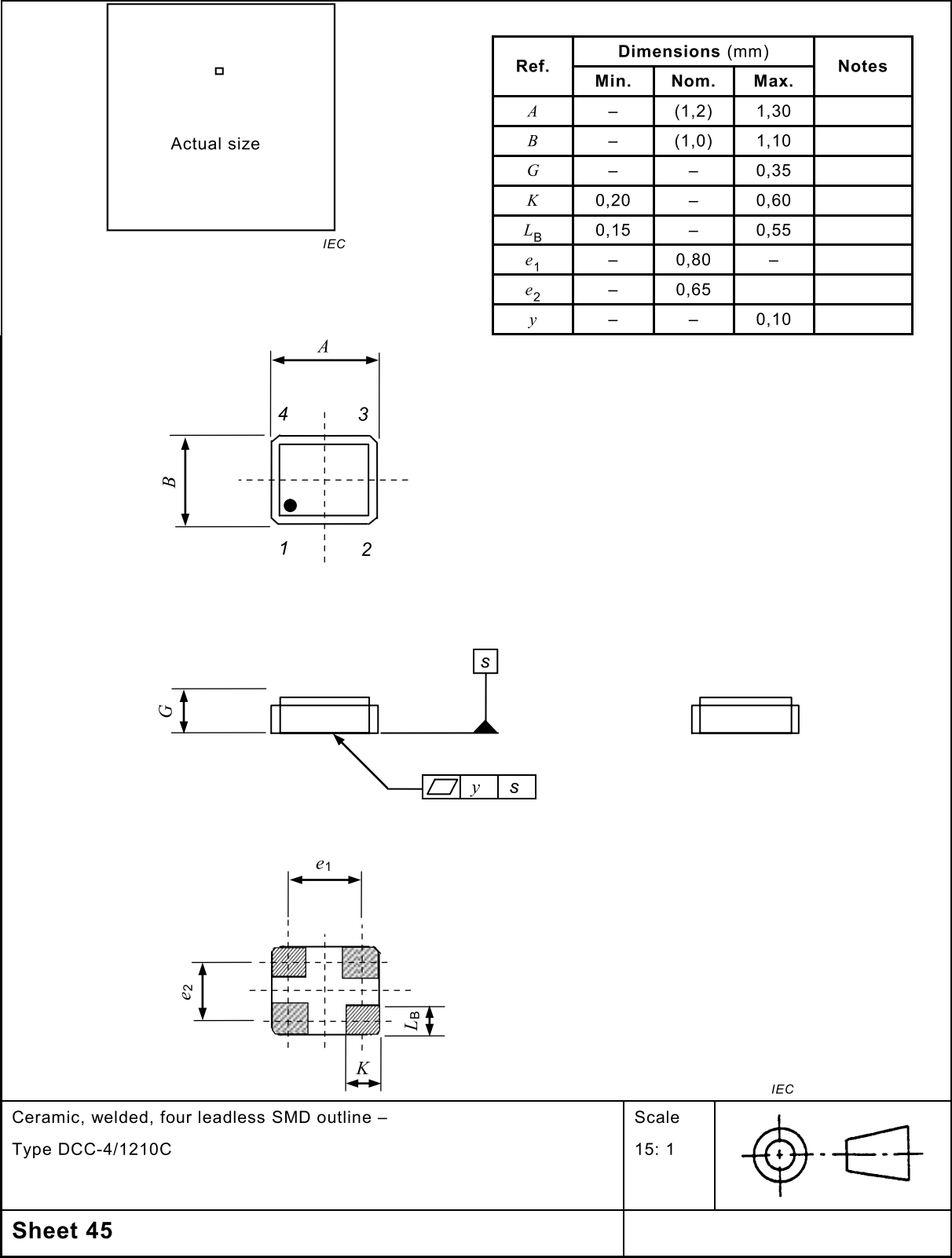
Terminal land connections of Type DCC-2/1612C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			



Terminal land connections of Type DCC-2/1610C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Terminal 2			



Terminal land connections of Type DCC-4/1210C

No.	Crystal unit	Crystal oscillator	Crystal filter	SAW devices
1	Terminal 1			
2	Ground			
3	Terminal 2			
4	NC			

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